
TITO FERRADANS
"ANAMORPHIC"
ON A BUDGET
2020 REVISED & EXPANDED EDITION
THE ORIGINAL ANAMORPHIC GUIDE!
THE WIDEST GUIDE ON SHOOTING DIY ANAMORPHIC!

DISCLAIMER: The original Anamorphic on a Budget guide is a result of its time. Things have changed, shops have closed, lots of new gear has flooded the market. Take this guide as-is. For the maximum and modern experience *check out my channel* and other guides!

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My graduation work at University of São Paulo in Brazil revolved around shooting with anamorphic lenses, Magic Lantern RAW and Canon cameras. The goal was to keep the budget as low as possible and maximize the experimenting.

The resulting document was written in Portuguese and I translated it in 2015 - *as I started my channel* - so more people could have access to my conclusions and would not take so much time going through the hit-your-head-against-the-wall research process.

This thing is around a hundred pages long, so it is thorough. I hope this work brings light onto common questions as well as supports your first steps into the world of anamorphic filmmaking today. Keep in mind this was written by December 2013 and 4K was not a big deal yet. To give you some reference, Blackmagic was just starting to ship out their first cameras.

I used a Canon 5D3 for this research and the camera is the least important part of it. Some of the sections have not aged well - I have not used Magic Lantern in a while, having switched to Sony in 2016, then Panasonic in 2019.

To encourage you to go through this amount of text in a quest for answers, here is the end result of my project, a two-part episode for a webseries, created by me (I wrote, directed, shot, edited and did all post-production). Both parts are 100% anamorphic and will be carefully analyzed over the following chapters. Be sure to enable English subtitles!

ZONA SSP (2013)



Episode 01



Episode 02

TABLE OF CONTENTS

I - WHO AM I AND WHAT IS THIS ABOUT?.....08

II - INTRODUCTION.....12

III - LENS RESEARCH.....17

IIIA – FIXED FOCUS (1.33X) 20

IIIB – ISCORAMAS (1.5X).....25

IIIC – DOUBLE FOCUS (2X).....29

IIID – CINE LENSES (2X).....36

IIIE - DAMAGE AND SERVICING.....42

ANAMORPHIC IN GENERAL.....44

ISCORAMAS.....45

LOMOS.....45

IIIF - DIOPTERS OR CLOSE-UP FILTERS.....46

IIIG - ANAMORPHIC LENS-YCLOPEDIA.....51

IIIH - FAKING THE LOOK.....52

IV – MAGICLANTERN RAW.....60

IVA – CUSTOM CROPMARKS.....62

V – ZONA SSP.....64

VA – LOMO – EPISODE 01.....66

VB – ISCORAMAS – EPISODE 02.....70

VI – THE PROCESS....73
VIA – CONCEPT DEVELOPMENT....73
VIB – ON SET PREVIEW....80
VIC - POST PRODUCTION WORKFLOW....82
VII - CONCLUSION....90
VIII - BIBLIOGRAPHY....92
ON ANAMORPHIC LENSES AND ADAPTERS....92
ON CAMERA WORK, LIGHTING AND HARDCORE DSLR SHOOTING..92
IX - WHAT LIES AHEAD....93
APPENDIX - ORIGINAL VIC - WORKFLOW....94



I – WHO AM I AND WHAT IS THIS ABOUT?

★ ★ ★ ★ ★

One of the main things for me in photography is the strong connection between technique and creativity one has to make to achieve good looking results. Knowing how the camera works is not enough. We need to understand human vision, composition, colors, light, and find/develop our own style.

When I enrolled in film school back in 2010 I was already taking pictures and studying photography for a year and a half. It was my main hobby and entertainment while not studying the subjects that would get me into university.

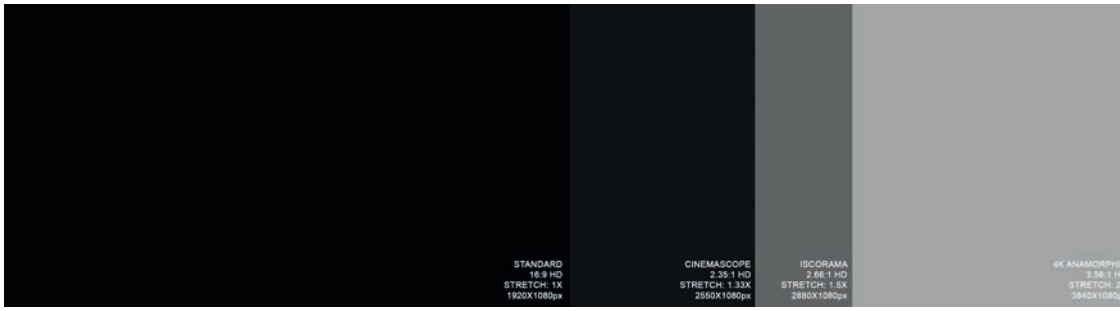
Coming from a math-based background (Computer Science at Federal University of Bahia), numbers were already exciting to me. Through my four years at film school I developed and emphasized the creative part of the photographic process as well - be it a single frame or 24 of them in a second.

During summer break in 2012 I was feeling bored and needed to learn something entirely new. I decided it would be an experimenting year, very different from the production-like work of the previous three years. Randomly roaming the web I came across a subject that had already caught my eye before: anamorphic lenses. It felt like the perfect time to go deeper into them. I was moving from 16:9 standards to something more movie-like as the 2.4:1 CinemaScope aspect ratio.

I had already come across the subject before around 2010 and read some articles about such lenses that would allow shooting CinemaScope straight from camera instead of cropping the frame. It was interesting but I lacked the technical knowledge to fully understand what was being said on those articles back then.

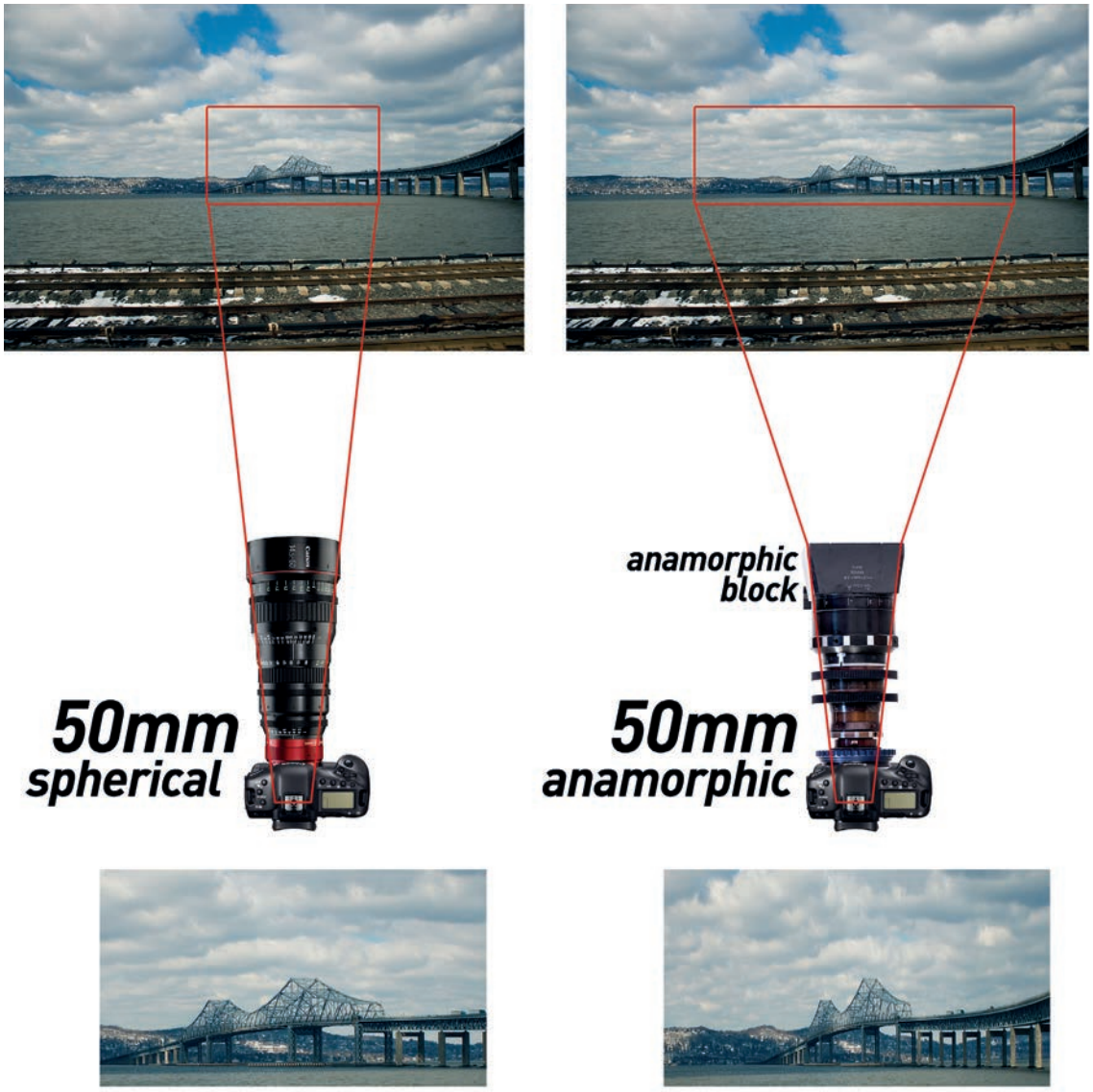
In the meantime whenever I considered shooting CinemaScope pictures, I always felt bad for discarding such a big part of the picture through cropping. Such a waste of money and production value, right?

Using the crop method for CinemaScope you either go ultra wide angle so you can frame your characters properly and end up with a ton of depth of field, or go with regular lenses and hope nothing too important ends up cropped off in post-production.



Without and with anamorphics aspect ratio comparison

After my initial readings I learned that anamorphic lenses squeeze the image before it reaches the sensor/film. Its horizontal correct measurement is determined by the image's default width multiplied by a stretch factor (which may vary from lens to lens). You do not discard parts of your frame but the opposite: you add to it! You bring in elements that would fall outside of a spherical lens' angle of view.


anamorphic
block
50mm
spherical
50mm
anamorphic

Since 2012 I read countless articles, threads, forums, posts, ebay auctions and even an ***e-book on the subject*** in order to better understand these lenses' inner workings. It is not all pretty and wonderful as it seems in the beginning but they achieve a great deal of what I was hoping for. There are various complications and quirks but nothing that is impossible to merge into film-set routine.

When I reached out for other anamorphic users in Brazil I did not find anyone, just a bunch of curious people like me. With the internet's connecting power I went virtually abroad and started researching and experimenting. During this process, way over twenty different anamorphic lenses went through my inspection and analysis.

This series of experiments were followed by ongoing posts on ***my personal blog*** aiming to introduce the subject in Portuguese. I was obsessed with these lenses and had to share my findings with whoever was interested. This feeling was particularly strong because I had a ton of questions in the beginning and could not find answers until I had the lenses in my hands, looking through the camera. I wanted to spare other people from that stress.

Research has been an amazing experience. During this process I sold all my spherical modern glass, keeping only vintage Russian lenses and anamorphics. I have no plan to leave them, I crossed the point of no return.

My main goal with this project is to share as much as I can of all my discoveries and tests with these unique and rare lenses, be it in terms of origins and history as well as practical use in real shoots.

I mean, amazing CinemaScope pictures in a camera that costs less than US$4k, when could I dream of this?

Author's note: Plenty of cameras these days qualify for that price range and we have a much broader variety in new anamorphic offerings! Things have definitely improved since 2012.

II – INTRODUCTION

★ ★ ★ ★ ★

What are these so called “anamorphics”?
The formal definition:

> **“Anamorphic; Producing, relating to, or marked by intentional distortion (as by unequal magnification along perpendicular axes) of an image.” - *Merriam-Webster Dictionary*.**

In our case, a lens or adapter that distorts only the image’s horizontal axis. This is called “stretch factor”. Each lens has its own and it ranges between 1.33x, 1.5x and 2x. That means the resulting image will be horizontally squeezed and in order to bring it back to normal proportion you need to unsqueeze (or stretch) it following these ratios. A different name for stretch would be pixel aspect ratio: a virtual number that defines the width of each single pixel when displayed (instead of the standard 1×1 Square ratio).

What is the point of having something that looks totally weird in camera and needs to be post-processed so it looks right? Anamorphic lenses were brought into cinema around 1950 when Twentieth Century Fox bought a patent for Henri Chrètien’s (a French scientist) system of bent lenses developed for astronomical research.

Years earlier, the film industry reached the conclusion that wider images are more appealing to human eyes. There was already a process to achieve such results. It was called Cinerama: movies were shot with three simultaneous cameras and showed in a similar setup with three projectors and a curved screen that was way bigger than the current standards. The downside was triplicating the use of celuloid for each movie during production and even larger amounts during distribution. That made Cinerama not as profitable as desired. But the audiences loved the wide screen aspect of it.

Around the same decade three different methods of anamorphosis competed in the film industry. The first one was based off prisms, developed and used by Panavision, the second was achieved through the use of bent mirrors by Technirama and the third one is our all-time favorite, cylindrical lenses that compress the captured image, originating the famous – and desired – CinemaScope. In numbers CinemaScope means 2.4:1 aspect ratio: for each height measurement, there is a 2.4x measurement in width.



Celuloid frame and different methods for a widescreen image.
Letterbox on top and anamorphic on the bottom.

In Chrètien's design the image compression would be directly related to the glass' curvature. His anamorphic lenses were made up by two optical blocks. The front block was responsible for the anamorphosis and the rear block was a regular spherical lens. They were both put together inside a single lens body. Simpler setups work the same way today (we will get to that) and you have to focus both spherical and anamorphic blocks separately.

From here on the spherical block will be referred as "taking lens". It is an important concept that will be used over and over.

Once the image is shot and squeezed onto the film roll – there was no digital intermediate back then! – another anamorphic lens was required for projecting the final motion picture in order to de-stretch the image onto the screen. These projection lenses are still out there, flooding eBay and every other online market. They are bulky, colorful and heavy. Another downside is their really distant minimum focus distance - since the screen and the projection booth is usually between 9 to 15m (27 to 45ft) apart.

Meanwhile the Cold War was going on and the USSR was also a major technological center. Across the world, other anamorphics were being made, Russian lenses manufactured by LOMO, the Soviet's main optical developer, responsible for all camera and movie gear in the country conglomerate.

These lenses were made solely for motion pictures and differently from the American/French design, anamorphic and spherical blocks were split by default. They could be connected through a simple bolt though, which synchronized focusing on both blocks, solving the issue of having to focus each lens individually. These were pretty strong builds and spread mainly across Asia. Just like Hollywood's standard, LOMOs had a 2x stretch factor.

At the beginning of the 1960's, ISCO Optics of Göttingen, German manufacturer, hires the man behind LOMO's designs. Deliberately replicating his previous work with LOMO, ISCO releases a series of anamorphic lenses, the Iscoramas. These had a stretch of 1.5x – aimed at the "rich amateur photographers" niche. The series consisted of a front anamorphic block attached to a 50mm f/2.8 cheaper taking lens (with Exakta, Minolta, Nikon F or Praktica M42 mounts). Image quality matched the market: sharp from center to corner.



The Original Iscorama – 1.5x stretch

In the 1980's Iscorama users found out it was possible to split both parts of the lens and use the anamorphic block paired to other taking lenses. ISCO takes advantage of this practice to release updated versions of their products getting rid of third-party manufacturers by discarding the 50mm taking lens and selling just the anamorphics. Those are the incredibly famous Iscoramas 36, 42 and 54.

The optical engineer behind these lenses used a design based on variable diopters. In this scenario the taking lens is focused to infinity and all the focusing is done on the Iscorama. By registering this patent, ISCO Optics of Göttingen killed any other manufacturer's dream of replicating its solution. This caused a drastic loss in terms of similar workarounds until recent times, making Iscoramas a unique species.

Quick recap: based on the need of creating wider-looking images while trying to NOT scrap every motion picture camera currently in use, scientists develop anamorphic lenses that attach on top of regular spherical lenses and compress their images. Russians are not far behind and develop their own anamorphics. Germans take the Russians' formula and improve it, releasing the Iscoramas which hold the single-focus patent.

Following these events, Japanese companies start making their own anamorphic elements (Kowa and Sankor) with a 2x stretch, and even Henri Chrètien, through Societe Technique Optique de Precision puts out some lenses for cameras and projectors, the Hypergonars S.T.O.P.

The market keeps evolving and while some of these lenses were pretty good with 8mm and 16mm film, with the change from film to video and later to all-digital formats, amateur anamorphics are less and less useful. They end up forgotten at the bottom of "used gear" boxes. Videocamera's lenses are not interchangeable and stills photographers were never too fond of the squeezed images. Around 2008 most of these anamorphic gems could be found on eBay for less than US$200.

The last additions to the anamorphics list came out after mini-DV cameras became popular. These are fixed focus adapters and their name comes from the fact that they lack a focus ring, featuring a simple screw for alignment – so the stretch is applied on the correct axis. Focusing is done on the taking lens, attached to the camera and it has its own limitations (more in ***IIIA - FIXED FOCUS (1.33x)***).

At the top of its game, Panasonic's DVX100 – the camera responsible for empowering the indie filmmaker – got its own anamorphic adapter, the Panasonic AG-LA7200, with much greater size and quality than its generic competitors.

The goal of these adapters however is not to achieve CinemaScope aspect ratio. Most mini-DV cameras (even the DVX100) shot 4:3 aspect ratio pictures, which was the default TV ratio at the time. These adapters came out to allow older cameras to shoot in the new and widescreen 16:9 "modern" proportion. This is much like what happened in the the film industry: it is cheaper to buy an adapter than a new camera. Unfortunately these were not a big hit because they were already pretty expensive and not many people were willing to pay the extra buck. They were also discontinued and forgotten.

We arrive at 2009 and all these lenses can be found on eBay for laughably low prices. The The Canon 5D MkII came out in the previous year and it is revolutionizing digital video's world. At that point a curious dude or gal, like you and I, decided to check if these old, weird and (at that moment) cheap lenses were compatible with DSLRs' video capabilities.

I can only imagine their surprise when the thing worked and, not only that, the resulting image was amazingly wide and unique. Results came online and more people got curious. Among these it is easy to name Eddie Enciu – responsible for Vid-Atlantic, an early adopter and a supporter of my research -, Edwin Lee – famous for being a pioneer in anamorphics and DSLRs – and Alan Doyle, also known as ***Redstan*** – a great source of knowledge about motion picture anamorphic lenses.

Shortly after processing the boost in width, these new anamorphic users realized a couple other features inherent to those lenses that were hard to achieve when not using them. These features had a lot to do with the resulting images' cinematic feel. The first of these features was the different bokeh or out-of-focus highlights, which remained oval even after unsqueezing. With spherical lenses bokeh has always more of a circular shape.

Another unique feature that quickly drove anamorphics up in popularity was the anamorphic lens flare. Lens flares are not always bad, especially if used while developing a unique look. Anamorphic flares are long, streaked and (popularly) blue. They are totally different from spherical flares and have a lot of personality. Using J.J. Abrams as a reference (easy one, I know), anamorphic flares are a constant through his work and we can easily spot his signature over a couple brief seconds in most of his movies.

From these various tests, reviews and experiments published on Y ouTube, Vimeo, and other online forums, many other shooters got interested in the game. So began the rush for anamorphic glass worldwide. In less then a year, Iscoramas that went for US$200 now reached US$4000 on eBay auctions. Many other lenses also sold for crazy-high prices.

Following this raise in value many anamorphic owners decided to sell their beat up old lenses, increasing the availability and lowering overall sales prices. Now,

a couple years after the initial outbreak, an Iscorama goes for about US$2500 on auctions.

LOMOs' prices are widely apart since many of them already have cinema mounts and quality standards, so they can easily go over US$4000, but it is important to keep in mind that these were never cheap and we are brought into the DSLR world through the use of adapters. I will not write about fixed focus adapters and projection lenses because their prices do not seem to follow as strict rules as the most popular Iscoramas and LOMOs.

Over the last months the anamorphic community rejoiced with announcements of brand new lens designs and releases to come soon. Among them we have SLR Magic – Hong Kong based – and Letus35 – in the US. Both promise 1.33x stretch and fixed focus adapters with controls for situations such as close up shots. Price expectations range between US$800 and US$1300.

III - LENS RESEARCH

★ ★ ★ ★ ★

After the first few weeks of research, reading, watching tons of test videos and getting to the core of the anamorphic squeeze's pros and cons, I still had plenty of questions unanswered. My curiosity got the best of me so I decided it was time to invest in buying lenses. My starting point was Reid's ***Anamorphic Shooter's Guide***, from there I went on refining my decisions and ended up buying four lenses, all of them quite different from each other. Their differences will be better explained along the following chapters. They were Kowa for Bell & Howell, Panasonic AG-LA7200, Hypergonar Hi-Fi 2 and a 50mm LOMO Squarefront.

Experienced shooters reinforced that I should use simpler taking lenses with less coatings to make flares pop. This meant no modern glass, no Canon L series, no zoom lenses. I ended up choosing vintage primes from 1970-80 and was pretty happy with Russian-designed M42 mount all manual (focus and aperture) lenses.

The kit included a 37mm f/2.8 (Mir-1B), 58mm f/2 (Helios 44), 85mm f/2.8 (Jupiter 9) and 135mm f/2.8 (Tair 11A). With these lenses I covered the standard focal lenghts in a basic prime kit and could work out my framing and composition with enough freedom to survive a shoot.

To bring out the anamorphic looks, it is good to pair scopes with vintage, pre-1970's, taking lenses, years before the development of multi-coating (MC). Multi-coating is a physical process of adding very thin layers of chemical

components that protect modern lenses from glass damage, reduce flaring effects and increase light transmission. This means that lenses made before this thing was invented are non-coated or single coated. That relates greatly to anamorphic's more proeminent feature: lens flares.

I picked mine in M42 mount out of personal taste but there are plenty of other mounts that can be easily adapted to Canon EF. Even though adapters are quite common, before buying any lens be sure to check if it works properly with the camera you are using. A simple search is good enough to save you from a headache involving flange distance and things being out of focus when they should be sharp. This also helps you to choose exactly which model you want and what features you wish to accentuate on your work. There are countless makes of 50mm around, which one suits you best?

There was still an important question unanswered: how to properly connect both anamorphic block in front and the taking lens behind it? LOMOs aside, all anamorphics can be considered adapters you add to the taking lens. They operate as a filter that you screw on the threads and deliver a very specific result.

In the very beginning of my searches I came across lots of posts stating that for a safe setup you should use clamps to connect both lenses. Clamps are metal rings that attach to the back of the anamorphic adapter - mostly using screws - and feature regular filter threads to screw onto your taking lens. This allows the closest possible distance between the two optics which saves us from the trouble of light loss when working with anamorphic adapters (usually the loss is so small it does not cost you even a third of a stop).



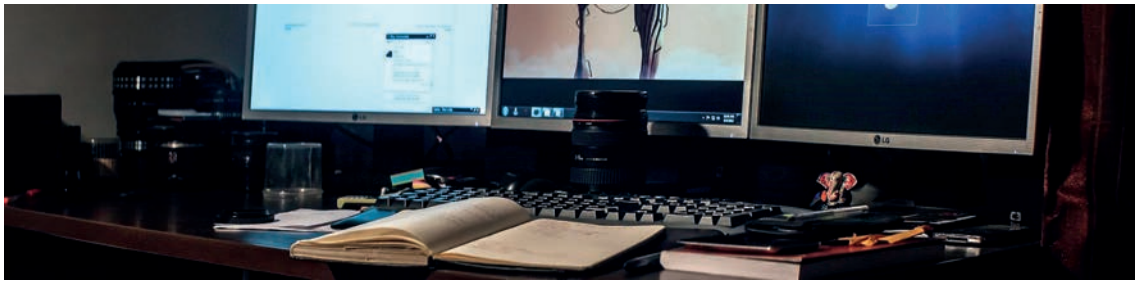
Redstan Clamp for Kowa Bell & Howell, 62mm threads

Many anamorphic adapters have threads on their backs already but these are not standard sizes and we usually end up with sizes that are not used nowadays.

Clamps serve to fix this issue as well as to deal with stretch alignment. You can loosen the screws without the adapter toppling over and rotate the anamorphic block so the stretch is properly aligned with the horizontal axis of the sensor. Screwing an anamorphic onto a taking lens and having it skewed in the camera is the first thing that happens when you switch lenses. It is fundamental to align scopes properly otherwise the final image will be irreversibly skewed. Check the pictures below for examples!



Misaligned anamorphic block



Aligned anamorphic block

The main advantage of getting a clamp made specifically for a lens model is that they fit like a glove, it feels solid and it makes for super easy alignment. There are very few manufacturers and a LOT of improvised workarounds. Redstan, in the UK, is quite famous for his specially-designed clamps of the most common lenses. ***Vid-Atlantic*** is an american company with cheaper options if you do not want to spend too much money on a piece of metal. Redstan's quality is unbeatable, but he is usually out of stock because the clamps take a while to cut and people are always buying them.

*Author's note: Since 2015 we have had **Rapido Technology** offer clamps and support systems for plenty of lenses, as well as **RAF Camera** for specific projects and smaller runs.*

As I said in the beginning of this chapter, each one of my lenses worked differently from the others. They had different stretches, image quality and years of manufacture. I will split them according to stretch factor because it is the easiest way to group and differentiate them.

IIIA – FIXED FOCUS (1.33X)

Author's note: This might be the biggest misconception from my original work and the main note in the revised edition. What I used to call "focus-through" lenses are a myth. These are actually fixed focus designs that, due to the lesser stretch, offer some tolerance for focus racking. Ultimately speaking, they are not set to close focus nor infinity, requiring modification for proper use with single focus solutions (variable strength diopters).

Just to be absolutely clear: there is no such thing as a focus-through lens in which all work is done with the taking lens and everything works marvelously. These guys have compromises in image quality.

The most subtle lenses when it comes to stretch factor are also the newest ones. Between 1999 and 2004 the standard screen size switched from 4:3 to 16:9 and, the same way it happened in the film industry, it would be too expensive to replace all the camera equipment currently in use – especially for those who had just bought an expensive mini-DV camera. So anamorphics were made to fit more screen size over less sensor size.

The adapters' manufacturers were scarce and their target market was composed by users of high-end cameras such as Sony PD170, VX2000, Panasonic DVX-100, Canon GL1 and GL2. They all shot 4:3 aspect ratio.

Panasonic made their own adapter, the AG-LA7200, one of the biggest 1.33x and expensive until this day. Sony and Canon were covered mainly by Century Optics - which ended up as part of Schneider Optics - and smaller brands that were less reliable when it comes to quality, such as Optex, Kenko, Soligor and others.



Optex 1.33x Adapter 16:9 Adapter

These cameras had very different filter mounts: Sony and Canon used different bayonet types on their high end models, others had regular filter screws and multiple sizes. Each adapter had a specific back for the camera it should work with. Another important fact is that most of them were only partial-zooms. This means they only worked well within a specific focal range on the camera's zoom range. From 50mm onward, the image would be completely blurry.

Focus was fixed somewhere between 4m (12ft) and infinity so there would be some serious blurriness issues when the subject was closer than that. This design's big advantage is that all focusing is done on the taking lens, with severe limitations for close focus or longer focal lengths.

These negative qualities – partial zoom and fixed focus – reflect on their modern use, paired with DSLRs. Lenses over 50mm and subjects closer than 4m (12ft) both require diopters to be in focus. Diopters, which also go by the name of close-up filters, are treated like filters but they are actually auxiliary lenses.

The biggest advantage of 1.33x adapters is that they allow the use of wide-angle lenses – between 28 and 35mm on a full frame sensor – without vignetting from the internal walls of the adapter. Besides that all the focusing is still done with the taking lenses, simplifying the shoot workflow.

Their build is simple, two cylindrical optical elements aligned with each other. All 1.33x adapters have a screw on the side. When loosened, this allows the user to rotate the glass and properly align the stretch direction.

Low quality on edges and corners is a common issue with 1.33x adapters. It is easy to notice strong chromatic aberration, loss of sharpness and barrel

distortion when the taking lens is a wide-angle at the edge of vignetting. Flares are also very strong, two thin lines easily recognizable by their strong blue tint. Very sci-fi and cool-looking.



Flare, lens distortion and chromatic aberration on a single picture.

As many of these adapters have bayonet mounts they require some modifications ***using step-up rings and pliers***. Another challenge is attaching diopters in front of them in a safe way since there are no threads for additional filters. There are clamps that work with the smaller models – Optex, Century Optics and Soligor – but the Panasonic AG-LA7200 is always a challenge due to its massive size.

Author's note: I addressed these issues in two different videos on my channel. ***This is how to make 3d-printed front clamps for any scope*** *and* ***this is for a custom designed filter holder for the LA7200***.



Century Optics with clamp for 72mm threaded filters.

The Panasonic's greatest advantage and reason it is still so valuable today is that it allows much wider shots than its competitors - 28mm lenses on a full frame sensor - without any vignetting. This and superior optical quality seal its place at the top over the smaller Century Optics and generic branded adapters.

The downside is its size, making a tough challenge of finding ***properly sized diopters***. You can use ***this link*** for a custom-made filter holder that allows you to use smaller diopters at the expense of vignetting.



Panasonic AG-LA7200 with the first version of my filter holder.

Without the filter holder the only diopter choices for this lens would be from the super rare Russian LOMO Foton-A, 105mm or 4.5" filters. These diopters are big enough to cover the Panasonic AG-LA7200 but they still require mods so they can be safely attached to the front of the lens.



LOMO Foton-A +1 diopter.

Near the end of the DV fever Century Optics released the only full-zoom adapter for this dying market. The WS-13 is a big lens (almost the same size as the Panasonic) that besides the alignment knob, also has a focus ring. This allows the user to have sharp images of subjects down to 0.5m (1' 7") opposed to the original 4m (12ft) for the partial zooms and also use taking lenses longer than 50mm.

Its performance is much superior compared to the other 1.33x adapters – even considering its double focus aspect. Double focus means you have to focus both taking lens and the adapter at the same distance. This adapter has a higher tolerance to double focus mistakes than 2x adapters, as we will see in the next few chapters. It is still possible to rack focus using only the taking lens as long as you plan the shot well and set the adapter's focus ring to cover the target range.



Century Optics WS-13-SB.

As you can see in the pictures of the adapters so far, the elements in these lenses are not circles but more of a square shape. This means that the oval shaped highlights – anamorphic bokeh – will not show up when this kind of adapter is used. First because 1.33x stretch is too subtle to be noticed on bokeh, second because the square aspect of the glass ***does not affect the light shape***. To achieve oval bokeh with 1.33x adapters it is necessary to use modified taking lenses (see ***IIIH – Faking the Look***).

On the bright side, when you have a 1.33x stretch applied to 16:9 footage the resulting image has a 2.36:1 height:width proportion which means you are getting CinemaScope aspect ratio straight out of the camera!

A problem that did not exist when these adapters first came out is that due to DSLRs large sensors, getting a sharp focused image using fixed focus adapters

can be challenging. Mini-DV cameras have 1/3" sensors which gave them a longer depth of field. When you pick a sensor that is almost seven times bigger than that, the best way to get a sharp image is to stop down the iris on the taking lens. Apertures between f/4 and f/5.6 are usually enough to achieve sharp focused results. Diopters also help a lot when it comes to improving the quality of the footage made with fixed focus adapters.

Author's note: Not all 1.33x adapters are fixed focus, and not all fixed focus adapters are 1.33x. Examples for the former case are the Century WS-13 I mentioned earlier, which is double focus, just like the Letus 1.33x AnamorphX Pro. For the latter, the baby Hypergonar is 1.75x, there are baby Möllers and baby Iscos that are 1.5x and several 2x projection scopes have fixed focus too.

I categorized 1.33x as fixed focus here because they are the majority of fixed focus adapters and also to make it easier to absorb these classifications. Anamorphics have lots of exceptions.

A brief comparison between pros and cons for fixed focus adapters:

PROS:
- All focusing is done on the taking lens;
- Lower prices than higher-end 1.5x and 2x adapters;
- Allows using wide-angle lenses;
- Strong flares;
- Natural CinemaScope.

CONS:
- Fixed focus limits your focusing range and requires modification for single focus solutions
- Strong lens distortion and chromatic aberration around the edges;
- Needs diopters when the taking lens is longer than 50mm;
- Needs diopters for subjects closer than 4m;
- Taking lens aperture should be between f/4 and f/5.6 for best performance;
- No oval bokeh.

IIIB – ISCORAMAS (1.5X)

Back in 2009 an Iscorama could be bought for little over a hundred bucks. Currently prices range between US$2500 and US$3500. It takes a lot of effort, patience and time to get good deals on these lenses.

Iscoramas were made between 1970 and 1980 by ISCO Optics of Göttingen, in Germany and work in a unique way. They are adapters in the sense that they fit in front of a taking lens and their high prices come from a great number of

qualities. There are three main models: 36, 42 and 54. Their naming comes from the diameter of the rear anamorphic element. The weight difference between the three is drastic. While the 36 weighs around 400g (1lb), the Iscorama 54 weighs 900g (2lbs).

Their stretch factor is 1.5x which leads to an aspect ratio of 2.66:1 for 16:9 sensors. That is slightly wider than the CinemaScope standard. 50mm is the widest you can go using an Iscorama on a full frame sensor while still avoiding vignetting. Anything longer than that is safe and clean.

While fixed focus and double focus systems have improved performance between f/4-5.6, Iscoramas shoot razor sharp images no matter the aperture on the taking lens. Their optical quality and performance is stellar.

Their biggest quality though is all focusing is done based on the Iscorama's focus markings. Focus your taking lens to infinity and then work just with the Iscorama, which has a minimum focus of 2m. It is exactly the same thing as shooting with a spherical lens and having the advantages of the anamorphic format. The way ***its focusing works is patented***, and no one broke that patent, so these are the only adapters with single focus.

A couple years after their original release ISCO noticed that their adapters were being paired with other lenses than the standard 50mm they came with. Upon that realization they took the chance to improve the glass and release them as standalone accessories. That is when the Iscoramas 36, 42 and 54 were born. They are more modern and multi-coated, so less prone to flaring than the original Pre36.

These lenses were designed for photography (not only projection) so they have standard filter threads on the front and back - 72/49mm, 82/62mm and 95/77mm for Iscoramas 36, 42 and 54 respectively.



Iscoramas Pre36 and 54, Iscomorphot 8/1.5x and Isco Widescreen 2000.

Their alignment mechanism is also simplified. There are buttons on the lens body that allow you to rotate the anamorphic block and align the stretch axis.

As any other anamorphics, Iscoramas also have exceptions. The first of them is the Iscomorphot 8/1.5x – also known as baby Iscorama. This is a version for 8mm film cameras which shares the same functionality as its bigger brothers: focusing is done only on the Iscomorphot while the taking lens is set to infinity. This little gem can focus down to 0.5m without diopters but its image has much lower sharpness and contrast when compared to its siblings. You start to get sharpness and contrast from f/5.6 or slower apertures. Combined with APS-C sensors, it is vignette free from 60mm on. Due to its tiny size, it does not work well with full frame cameras.



Iscomorphots 8/1.5x duo.

The second exception to the Iscorama family is the Isco Widescreen 2000 which is a fixed focus adapter with improved optics and 1.5x stretch. With this one, focusing is done on the taking lens and the adapter is fixed between 4m and infinity, requiring diopters for closer shots. It was used as a projection lens, so it needs clamps and mods in order to provide standard filter threads.



Isco Widescreen 2000 and accessories.

Not taking these two exceptions into account, here is the pros and cons list for Iscoramas:

PROS:

- Single focus!
- Simplified alignment;
- No need for clamps;
- Strong flares (only with the Iscorama pre36);
- Subtle oval bokeh;
- Great sharpness even at fast apertures;
- 1.5x stretch leading to footage close to CinemaScope aspect ratio;
- Regular threads for diopters and filters.

CONS:

- VERY high prices;
- Minimum focus of 2m (can be solved through diopters);
- Multi-coated (Iscoramas 36, 42 and 54).

About the Iscorama 36: since the lens' body is made of plastic, it is quite common that it gets damaged. Aiming at this valuable lenses market, British company ***Van Diemen*** developed a new housing for the Iscorama. The new body is fully made of metal, featuring follow focus gears and a minimum focus modification that drops it to 1.2m instead of the original 2m.

The process costs £850 and has an estimated turnaround of 90 days plus mailing time. The conversion's ***results are impressive*** but there are a lot of customer complaints about waiting a much longer time than advertised to get their lenses back.

There is also a homemade modification to improve the lens' minimum focus by unscrewing the front element and removing the original focus stopper. It is risky if you are not used to opening lenses, but works perfectly.

I sent my Iscorama 36 for Van Diemen's rehousing and ***wrote a full review of the process***.

*Author's note: Since late 2017 the **Proxiscope/Maxiscope** mod is also an option when it comes to modifying Iscoramas 36/pre-36. These offer improved minimum focus, focus gears, rail support, alignment adjustments and a full metal body. It costs a fraction of Van Diemen's fees, plus the process can be easily reversed. The only issue is batches are scarce.*

IIIC – DOUBLE FOCUS (2X)

*Author's note: This guide was written way before the announcement of single focus solutions or variable diopters such as the FM Lens, Rangefinder, Rectilux and Rapido FVD, among others. **Here is a breakdown on how they work**. Variable strength diopters brought new life to double focus scopes.*

Entry point for many anamorphic users, these lenses are cheaper because they present a lot of challenges to work with. All of them are projection lenses, have an awesome 2x stretch and are placed on top of a taking lens. As they are projection lenses and the screen is NEVER close to the projection booth, it is quite common that their minimum focus ranges between 6 to 9m (18 to 27ft).

The name of the chapter introduces one of their challenges: double focus. In order to have a sharp image you are required to focus both the taking lens and the anamorphic block to the exact same distance. If one of them is slightly off, image quality degrades quickly.

Taking this into account, it is easy to figure that camera movements and rack focusing are very complex operations to be executed using one of these lenses. Your shooting style has to be adjusted to fit the lenses.

It is a practical thing to split them in two main categories: 35mm projector lenses – large, heavy and sometimes colorful – and smaller projector lenses like 8mm or 16mm, used in airplanes, military applications or film clubs.

Some 35mm anamorphic projector lenses are still in production by Schneider Optics. There are also vintage models made by Isco, Kowa, Sankor, Bausch & Lomb, Carl Zeiss and Möller, among a few others. The main advantages of these 35mm projection lenses are high image quality and sharness from edge to edge of the frame. Because of their minimum focus distance (longer than 6m/18ft), diopters are key parts to using these lenses in narrative projects. Besides, due to their weight, a ***proper support system*** is required.

The taking lens has to be considerably longer than the ones used in the previous examples. 85mm is the widest you can go on a full frame sensor to avoid side vignetting. Working out the math for focal length equivalent, this comes close to a 45mm on the horizontal axis.



Henri Chrètien Hypergonar Hi-Fi 2 and its support system.

Another obstacle: none of these lenses have filter threads, preventing proper use of diopters to improve minimum focus. Some have clamps to hold filters, but in most of them you will have to improvise a solution. For example, I glued a 95-86mm step down ring to the front of the Hypergonar Hi-Fi 2, allowing the use of filters with both sizes.



95-86mm step down ring glued to the front of the Hypergonar Hi-Fi 2.

Getting to the more interesting part of this chapter: lenses for smaller projectors. Manufactured almost exclusively in Japan by Kowa and Sankor during 1970-80 and shipped out to the whole world, these lenses come in different shapes and sizes but their mechanics and performance are quite similar. Minimum focus sits at 1.5m/5ft and there are lots of properly designed clamps and rings to attach filters onto either ***Kowa*** or ***Sankor***.

The most wanted versions are Kowa 8-Z and Sankor 16-D, which used to be the high-end versions of the smaller ones. It is quite common to find identical lenses with different labels – Eiki, Elmo, Vidoscope, D.O. Industries, Singer, the list keeps going – but their manufacturer was always the same, it is just the name that changes.

Author's note: After years of misconceptions, I finally stacked all the Kowas against each other to prove that the 8-Z, 16-H, B&H and Elmoscope II use the same glass. It is also claimed that the Sankor AnamoPrime and Sankor Compact share the design but I have not tested them personally.



Sankor 16-D plus clamps and accessories.

The 2x stretch makes anamorphic bokeh very pronounced when using these lenses, but 16:9 video ends up with an absurd 3.56:1 aspect ratio – a little image strip of image in the middle of a black screen. The best workaround for this issue is to shoot 4:3 as RED cameras in Anamorphic mode, or Alexa 4:3. Canon DSLRs through Magic Lantern and even the Panasonic GH4 offer such crops. This way the product results in 2.66:1 aspect ratio.



Strong anamorphic flares and bokeh.

One of the differences between 35mm projector adapters and the smaller ones is the coating used on the glass. Most 35mm are multi-coated since they are modern lenses and this kills anamorphic flares. On the other hand, the smaller ones are single-coated and present various colors of flares that cannot be achieved in any other way.

One of the most dreaded features introduced by projection lenses are "anamorphic mumps". These first came into existence when Hollywood started using anamorphics and it is still a common feature among the vintage scopes. David Bordwell explains it way better than I ever could:

> **"(…) the 'squeeze ratio' of Chrétien's lens design varied across the horizontal axis. These optical tics created distortions and patches of soft focus. The most embarrassing flaw, created by faults in magnification and the uneven compression of the visual field, made central figures look oddly bloated. In close ups, the result was 'CinemaScope mumps'. Not all of the films were shot with Bausch & Lomb lenses, but other brands of anamorphic lenses tended to cause the same problems." (BORDWELL, 2007, p.288)**

As Bordwell describes, the cause of this quirk is the uneven glass curvature that creates the stretch factor. Around the edges of the lens compression is stronger while in the middle the image tends to be more spherical-like.

Stretch is not constant across the horizontal axis and it also changes according to the focus distance. For objects that are further away and closer to infinity, the stretch is 2x, but when the subjects come closer to the camera, this number can go as low as 1.7x. When this footage is stretched back to its "proper" aspect ratio whatever subject is in the center of the frame becomes wider than it should be, thus creating mumps.



Mumps on a close up shot using a 135mm taking lens + Kowa for Bell & Howell.

There are also exceptions and special cases such as double focus lenses with 1.5x stretch designed for amateur 8mm and 16mm cameras such as Bolex-Möller or Yashicascope. Because of their size they have been kindly named baby anamorphics. As they exist in smaller numbers and from different manufacturers their optical quality varies greatly.

Taking these exceptions into account, the few double focus adapters that can be compared to Iscoramas when it comes to image quality are Kowa for Bell and Howell (and others, as my previous note explains) and Bolex-Möller. The first ones, from Japan, were made specifically for an European projector company (Bell & Howell) that demanded extensive quality tests, so this brand of Kowas can be set apart from the other ones.



Kowa for Bell & Howell with Redstan clamps.

Bolex-Möller are Germans and, as the name indicates, come from a partnership between the camera maker Bolex and Möller optics, which means these are lenses for shooting and not projecting. Their minimum focus is at 0.5m/2ft. There are two different versions of Bolex-Möllers: one for 8mm and another considerably bigger for 16mm. After a while, the partnership ended but Möller still made some more anamorphics on their own. Even though the markings are different, the optical quality is identical.



Bolex-Möller 8/19/1.5x.

2x stretch is only achievable through thicker glass elements. Because of this extra thickness another problem comes to life: while using DSLRs, depth of field can be so shallow that it fails to cover the whole thickness of the anamorphic glass. In this situation, the center of the image is in focus, but there is a mismatch between vertical and horizontal fields on the edges of the frame, reducing image quality.

Taking this into account, in order to shoot high-end footage it is better to work with smaller apertures, granting enough depth of field to cover not only the center of the adapter. Apertures between f/4-5.6 tend to have excellent results.

PROS:

- Ideal anamorphic bokeh;
- Lower prices than other anamorphic adapters;
- Minimum focus between 0.5m and 1.5m (for older lenses);
- Strong flares (for older lenses);
- Great image quality from edge to edge (for modern lenses).

CONS:

- Double focus: it is necessary to focus both the taking lens and the anamorphic at the same distance so the image comes out sharp;
- Big and heavy (for several lenses);
- No filter threads;
- Very long minimum focus (for 35mm lenses);
- Taking lens should be stopped down for better performance;
- Camera should be fixed because rack focusing is a nightmare;
- Mumps and distortions across the frame;
- 3.56:1 aspect ratio.

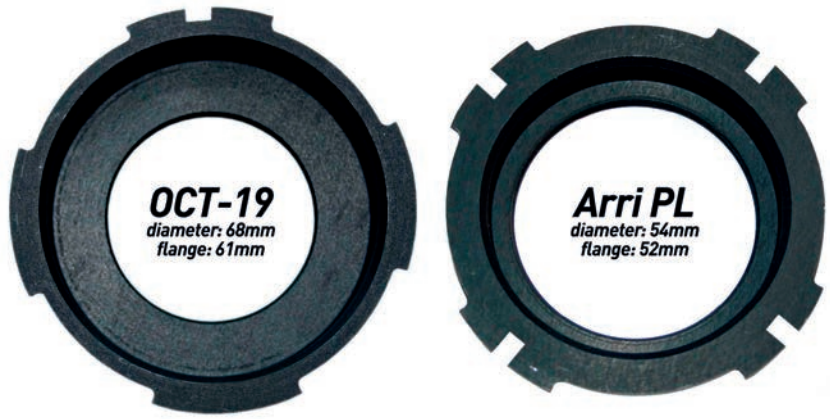
IIID – CINE LENSES (2X)

The best was saved for last. Anamorphic Cine Lenses still exist in great numbers, and by 2012-2013 they were being announced by many major optics developers (***Cooke***, ***Zeiss***, ***Angènieux*** and ***Hawk***) for cameras like RED, Panavision and Alexa. Sadly for this work, a single cinema lens costs tens of thousands of dollars and falls outside the definition of "affordable prices".

As presented in ***II – Introduction***, the cinema anamorphic lens goal is to double the amount of image captured on film so there is not much math required to figure their stretch of 2x. Currently Hawk makes some with 1.33x stretch but all others (old and new) have a stretch factor of 2x. These lenses have no quirks and exceptions when it comes to usable aperture and close focus.

Among the best options at low prices (between US$800 and US$4000) are LOMOs, designed and manufactured at the technological apex of the Soviet Union. There are two main series of prime lenses: Squarefronts and Roundfronts - their names hint at how they look. Most common mounts are OCT-18 and OCT-19.

OCT-18 is the oldest of the two and it is the Russian version for the international Arri Standard Mount, just a tad bigger. Lenses with this mount go deep into the camera and twist inside the mount itself due to a locking pin that acts as focusing mechanism. This results in a very short distance between the rear glass and the camera sensor, preventing their use in cameras that have mirrors inside and working properly only with mirrorless models like Panasonic's GH series, BlackMagic Design's and many of the Sony Alpha series.



Arri PL and OCT-19 side by side comparison.

OCT-19 mount is also the Russian version for another international standard, Arri PL, a little larger as well. It replaced OCT-18 for being sturdier and easier to handle.

The curious part of this story is that both mounts (OCT-18 and OCT-19) were developed from cameras captured during World War II and redesigned so Russian lenses would not work in cameras other than Soviet Union's.

It is fairly easy to find lenses with both kinds of mounts. OCT-18's go for lower prices, but either is much cheaper than any non-Russian anamorphic cinema lens.

Squarefronts are older, from 1960-70, and their look is unmistakable. Their front is square. Every lens features the acronym NAS (HAC in cyrillic) which stands for anamorphic attachment. These lenses are composed by two distinct optical blocks: a spherical one and an anamorphic one, similar to the many other options presented in this guide.

The Soviets' greatest innovation was a combined focus system in which a bolt connects the two optical blocks and the focus scale is absolutely identical on both. Because of this, when focusing with the anamorphic block the user automatically adjusts the spherical block's focus as well. It is a very simple solution and beats every other idea so far in terms of efficiency.

The big issue from having two separate blocks is the need for a proper support system so the anamorphic block will not topple and crash on the ground.





LOMO 35HAC4-1 50mm T/2,5 OCT-18.

An interesting aspect of these designs is that some of the anamorphic fronts are compatible with more than one spherical back. The most common case is having a single anamorphic front for 75mm and 50mm backs, lowering the cost for a complete set and also making your camera bag lighter.

On the following decades came the Roundfronts, BAS (БАС, in cyrillic) acronym, meaning anamorphic block: the lens does not split in half, it is a single unit containing both spherical and anamorphic blocks and a single focus ring. It is an anamorphic lens that works as… a regular lens! Without quirks. Because of that they reach much higher prices than their Squarefront sisters. Their mount is usually OCT-19 or Arri PL when upgraded to work with modern cameras. Many professional cinematographers like using Roundfront sets nowadays because they have a very different look when compared to modern anamorphic lenses, which tend to be a little lacking in personality.



Squarefront and Roundfront LOMOs side by side.

Among these prime series LOMO also developed a couple zoom lenses, harder to find and pricier to buy. Between these, the easiest to afford is the Foton-A, which works with Squarefront's combined focus mechanism and can be split in spherical and anamorphic blocks.



Foton-A 37-140mm T/4.4 OCT-19 split anamorphic and spherical blocks and +1 diopter.

There are also single block zooms, like Roundfronts, but these are almost impossible to be found for sale. When they show up, prices are always above US$10.000.

In the 1980's, outside of the USSR, rear anamorphic adapters were born. These were designed to work with a very specific group of lenses: super-zooms. These ranged from 20-25mm on the wide end up to 100-250mm on the long side. The most popular ones were Angènieux's and Cooke's, which shared very similar designs. The adapter was to be attached behind the last optical element of the lens and squeezed the image right before it reached the film.



Shiga Rear Anamorphic.

Russians would not stay behind as they had lenses "inspired" by the same designs from Cooke and Angenieux. They soon made their own rear adapters. Aimed at a few LOMOs, each adapter only worked with a specific super-zoom.



Rear anamorphic adapter for LOMO 350PF-18-1 20-120mm T/3.3 OCT-19.

As these adapters came between the back of the lens and the film strip, they automatically caused some light loss – usually 1 stop – and an "increase" in image size, in similar effect to a tele-converter.

The easiest way to wrap your head around this idea is to imagine the whole thing as a projection system: If the screen is close to the projector, your image is brighter and smaller. As you increase the distance between them, the image becomes bigger, darker and some of it falls off the screen. So, when this kind of adapter is installed, the focal length of the lens doubles. A 25-250mm lens becomes 50-500mm, a 20-120mm becomes 40-240mm and so on. The greatest advantage here is that even though the focal length has been doubled, due to the 2x squeeze, this increase is only noticeable on the vertical axis. It is confusing to explain this with words, so I am summoning images to help me right below.



Spherical 20mm.



Spherical 40mm, 2x tele-converter.



40mm with rear anamorphic adapter.

It is easy to see that even with a longer lens (40mm compared to 20mm) the image loss happens only in the vertical axis. With these little accessories the need to make anamorphic super-zooms was dead, providing a quick way to swap between a spherical and anamorphic version of a single lens.

A rear adapter gets rid of flares and oval bokeh which are some of the most desired anamorphic features – also despised by most DOPs during the 1970-80s – so a lot of folks today would not be interested in rear adapters. For those who want to shoot scope as clean as possible and optimize sensor use, rear adapters are the way to go.

While researching cine anamorphics in Brazil I had lots of help and support from my friend Gustavo Fattori. Thanks to him we took a look at a Hawk set in a local rental house (Rentalcine, São Paulo). We used a RED Epic for testing. The set comprised 35mm, 50mm, 75mm, 100mm and 135mm focal lengths, marked as Hawk B-Series with 2x stretch.

The 35mm is a behemoth: the front glass is almost the size of a tea plate and the lens weighs over 3kg/6.5lbs. Edge distortion is very strong as a wide-angle lens usually does, but even more prominent due to the bent glass. After this one, we tried the 50mm which had a loose element inside the lens and required servicing. Lens flare, in both cases, was not too fancy and both were also very dark although marked as T/2.2. We ended up not trying the other focal lengths.



RED Epic (5K Ana) + Hawk B-Series 35mm T/2.2 Anamorphic.

A curious piece of information regarding this set is the “B-Series” naming. Hawk’s official website only mentions C-Series and V-Series. I found ***one single reference*** to the name saying this is a less reliable and cheaper design developed exclusively for the Indian cinema market and not made in great numbers or anywhere else in the world. It got me wondering how an Indian cine lens kit ended up in Brazil. What kind of deals this lens case went through?

Even defective, dark and questionable in terms of quality, these are still much superior to projection lenses, mini-DV adapters, Iscoramas and even LOMO Squarefronts. Its quality is matched by Roundfronts, but falls behind when compared to the anamorphics Hollywood currently uses.

IIIE – DAMAGE AND SERVICING

All anamorphics taken into account in this research are no longer in the making. The last batch for the digital video market was discontinued in the early 2000's. Most other lenses come from 8mm and 16mm projectors and cameras sometime between 1960 and 1980. Due to the fact that people are so curious about these lenses and all the weirdness involved in putting them to use, a great number of scopes go through several owners in a short amount of time. There are also those which were kept stored for decades in inadequate conditions and suffered with humidity, heat and fungus.

A lens can present a multitude of problems. The most common ones are scratches and fungus but there is also balsam separation when the cement between glass blocks grows weaker and they start to fall apart from one another, cleaning marks which can ruin image contrast, and many others: grease can become stiff, tightening focus rings, threads might crack, strip or bend, screws and bolts get lost, plastic breaks, you get the idea…

It is quite common that a recently purchased anamorphic requires some love. In many cases the lens is still working alright but servicing brings its quality up sensibly.

When I bought my LOMO Foton-A, it had to go through a lot of mods - mount replacement, declicking aperture, general cleaning and gear installing. I chose Olexandr Kalynychenko for the job. He is a specialist in Russian cinema lenses based in Ukraine. Communication was in English and Olex is always very clear with information and suggestions, sending pictures and explaining every single detail including the costs for each operation - plus if it is a good investment or not.

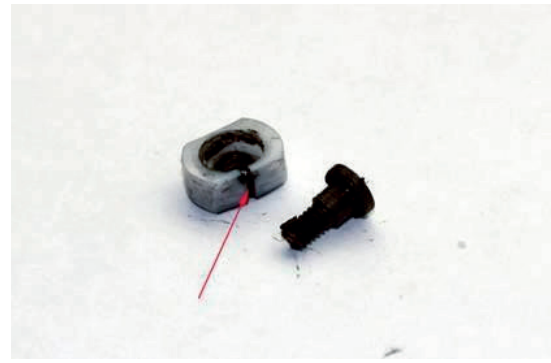
Using myself as an example, we talked a lot about how to adapt the anamorphic front support to 15mm rails (light rig standard). During this process I sent him ideas and he gave me his input. We ended up with a solution so good that he decided to make one for himself too and offer it to future customers.



LOMO Foton-A with custom 15mm rails support.

During the servicing process he also sent me pictures of the disassembled lens pointing out a small plastic piece that was partially cracked. He said the lens would still work as it was but the thing could snap at any moment. He offered to make me a new one and gave me his price. I decided not to take chances with a small plastic piece messing up the lens and asked him to replace it. These are the pictures he sent me.





The Foton-A was not my only lens that gave me trouble. My original Iscorama came with its front glass pretty messed up. After some questions and tips on various online forums I shipped the lens to John Van Stelten in Colorado, in the US. John is the guy in charge of Focal Point Lens and before ISCO closed its doors he was officially the man responsible for all fixes and repairs on ISCO glass in the US.

Communication was quick. We exchanged a couple messages and I showed him the problem on various photos. He explained me his theory about the damage and gave me instructions on how to check how serious it was as well as a price for polishing the external surface of the glass and recoating. He said he would only be sure of what was needed once the lens got to his hands. On his message after receiving the package and taking a closer look at it he pointed out other problems that he could not fix: a deeper scratch in the rear element and the torn off focus stopper (explained in ***IIIB – Iscoramas (1.5x)***). There are pictures to compare the lens before and after servicing in ***VB – Iscoramas – Episode 02***.

In both cases – LOMO and Iscorama – servicing was not cheap, but way cheaper than trying to find a new, perfect model of the same lenses. If you have a defective lens and it did cost you some serious cash, trying to fix it is a much better option than just giving up hope.

In order to make this task easier I put together a list with reputable lens doctors.

Author's note: In recent times (2018/19) I heard really scary stories from both Van Diemen and Super 16, Inc, which make me weary of sending them any gear. I also had an excellent experience with Ilya Volkov at Moscow Optical Workshop.

ANAMORPHIC IN GENERAL

Super 16, Inc (New York, USA) – Bernie O'Doherty fixes and collimates anamorphic lenses. According to him, a slight 1/64th to 1/32th of a degree off in alignment can mean the difference between a perfect image and one that needs f/4-5.6 to look ok. And customers ares used to this low quality when it comes to anamorphics! Of course, some lenses have design flaws that prevent perfect images, but the great majority of lenses – anamorphics included – can benefit from good servicing. Redstan always says wonders about Bernie too, that he fixes lenses since the 70s.

contact info: *bernie@super16inc.com*

ISCORAMAS

Focal Point Lens (Colorado, USA) - **CLOSED** – John Van Stelten was the official tech assistance for ISCO in the United States until ISCO was bought by Schneider. John has seen a lot of Iscoramas. He is one of the few techs who offer polishing and recoating. It is not cheap and it can be risky but in many cases it is the only way to eliminate all damage to the coatings or the glass.

contact info: *john@focalpointlens.com*

LOMOS

Olex Camera Services (Ukraine) – Olex is amazing. His messages are always clear and show that he A LOT about any lens you might need some more information. If you have questions, shoot him an email, I am pretty sure he will tell you what you need to know, and quickly.

contact info: *olex.camera@gmail.com*

RafCamera (Bielorussia) – Rafael is a businessman. When LOMOs were first coming to the DSLR market he was the first one to grab the opportunity and sell a great deal of lenses and accessories. He currently still sells some old accessories and makes custom gear for LOMOs and scopes in general. He has a good reputation on eBay and can provide you with some ideas of where to look for specific parts for lenses.

contact info: *support@westernbid.com*

One thing is key when walking the path of servicing you lenses: patience. Every piece is fagile and the adjustments are delicate. If you pick a decent tech to fix your lens, they are likely to have a queue before you because of a quality of work that many other people want.

IIIF - DIOPTERS OR CLOSE-UP FILTERS

For best results when working with anamorphic adapters, the constant use of close-up filters is required. Because of that, many anamorphics need some modification to their front part since they lack standard filter threads or any filter threads at all. These modifications are front clamps. They tend to be very specific to each anamorphic model and usually have 58mm or 72mm filter threads. Step rings are a simple and cheap way to allow for other filter sizes.



Redstan front clamp for Kowa Bell & Howell, 72mm filter thread.

Minimum focus is around 2m/7ft for the vast majority of lenses. 1.5m/5ft in some cases, which is still far to shoot a close up or framings tighter than that. Besides, when using an aperture faster than f/4, focus can be a challenge of its own.

Diopters – or close-up filters – are actually auxiliary lenses consisting of a single optical element or two (achromatic doublets or achromatic diopters), mounted on standard filter threads. They are named after their focal range, measured in ***diopters*** (to make it more understandable, the "diopters" measurement form will be called "power" from now on). Powers range from +0.25 up to +10 (the + signal means they are converging lenses).

A diopter is always placed at the very front of a lens/optical system and it limits the distance represented by infinity focus. The "new infinity" is much closer and stands for the farthest focusable distance from the front of the lens. This can be determined according to the power of the diopter.

The math is quite simple but we first need to define two variables. All measurements here are in meters.

MaxF = farthest focusable distance, in meters.
S = diopter's power

$$MaxF = \frac{1}{S}$$

Taking +0.5, +1.25 and +2 power diopters as examples, we have:

$$MaxF = \frac{1}{0.5} = \frac{1}{\frac{1}{2}} = 1\ x\ \frac{2}{1} = 2m$$

$$MaxF = \frac{1}{0.25} = \frac{1}{\frac{5}{4}} = 1\ x\ \frac{4}{5} = 0.8m$$

$$MaxF = \frac{1}{2} = 0.5m$$

As power increases, maximum focus distance shortens. In a practical way, how do these lenses improve anamorphics performance? Anamorphic setups perform best when focused closer to infinity, losing quality as the subject comes closer to the minimum focus mark between 1.5m and 2m. If infinity is now limited to a couple meters or centimeters in front of the lens because of the diopter, excellent results are now achievable in this area that used to be minimum focus.

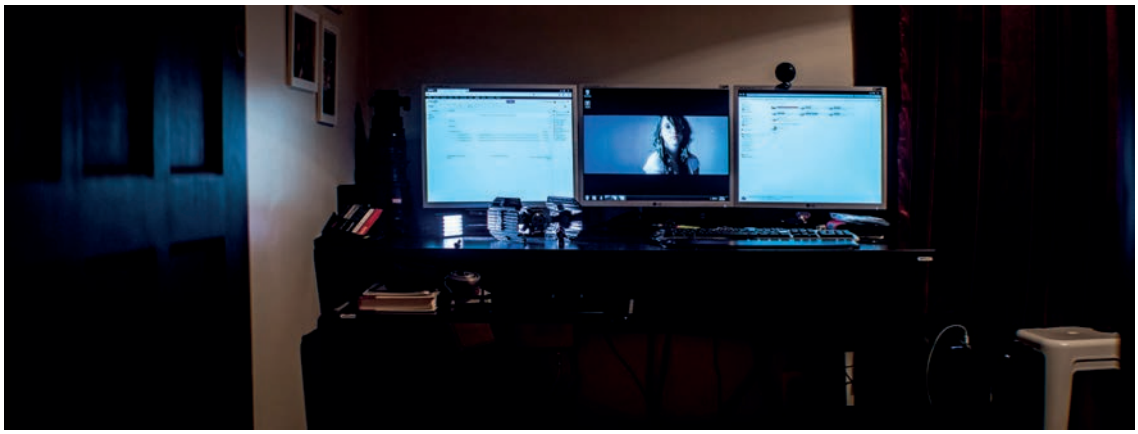
There are two obvious consequences to using diopters. The first one is that it becomes impossible to focus anything further than their farthest distance. The second is that every single focus mark on the lens barrel is now wrong. So ignore them. It is faster and easier to eyeball focus when using diopters.

There is an operation to determine precisely the minimum focus of a lens when attached to any diopter. This operation can also be used backwards in order to determine the strength of an auxiliary lens if you know the lens minimum focus with and without the diopter – particularly handy for poorly written eBay listings or buying weird zoom lenses diopters that don't have their strength explicitly written on the side.

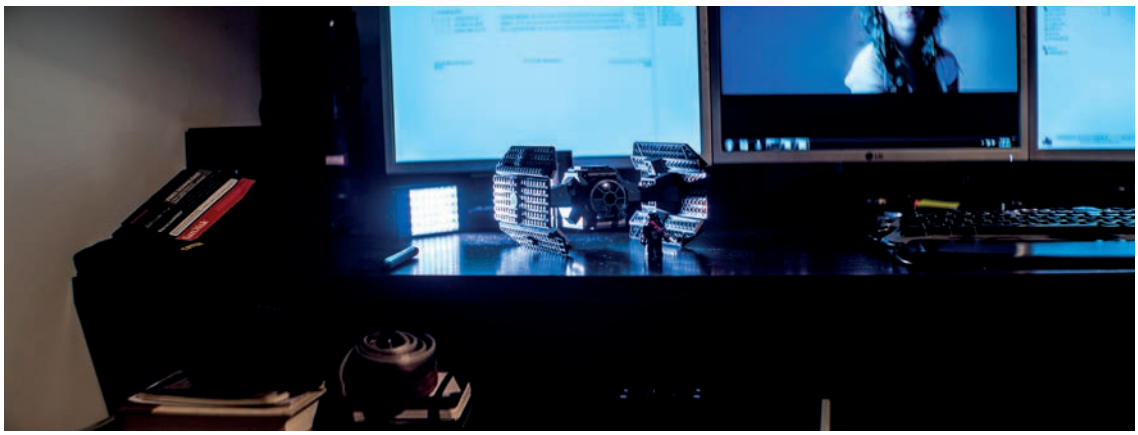
X = lens minimum focus without the diopter, in meters
X' = lens minimum focus with the diopter, in meters
S = diopter strength

$$X' = \frac{x}{(sx + 1)} = 0.5m$$

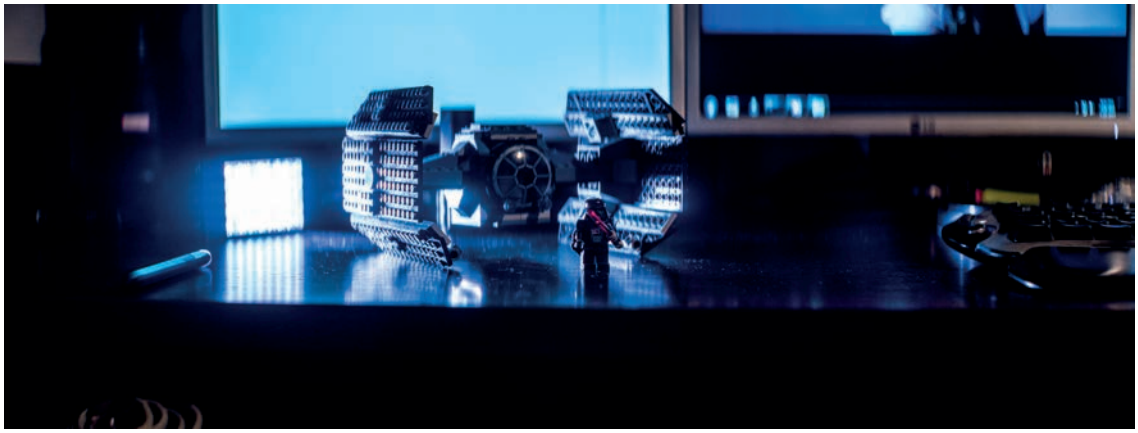
Shall we look at some examples? Pay close attention to the defocused objects in the background. On the first image, almost everything is in focus, but in the last one, the icons on the monitor are barely recognizable.



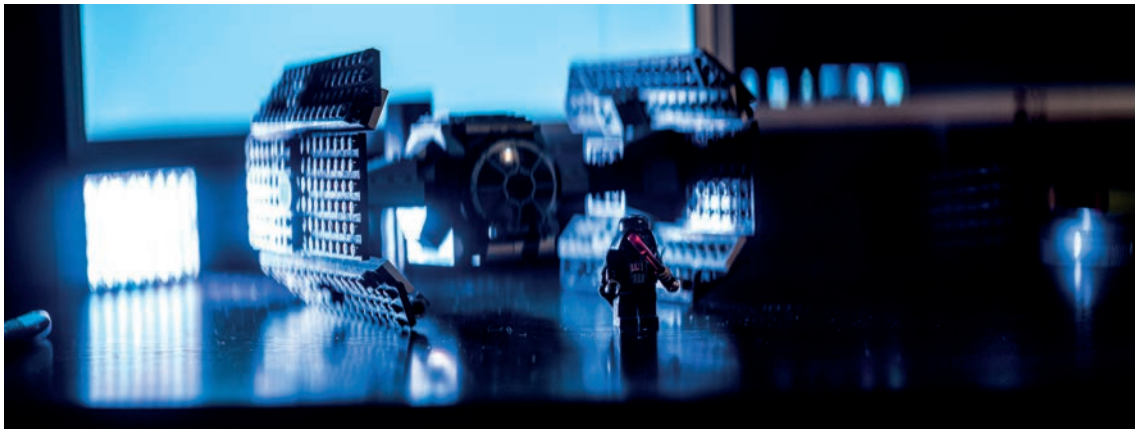
50mm taking lens + Iscorama 54 @ minimum focus: 2m.



50mm taking lens + Iscorama 54 + +0.5 diopter @ minimum focus: 1m.



50mm taking lens + Iscorama 54 + +1 diopter @ minimum focus: 0.5m.



50mm taking lens + Iscorama 54 + +2 diopter @ minimum focus: 0.25m.

eBay is a great source for diopters of all sizes. It is just a matter of looking for the most appropriate options to the scope you are using. 72mm thread is the standard for the smaller lenses like Century, Kowa, Optex, Sankor, Iscorama 36. Bigger lenses such as the Iscorama 54 or most projection lenses require bigger diopters and those are way less common, hence pricier too.

Do not get attached to any particular brand. Most of them have some sort of “magical” numbers to inform the diopter’s threads and power - and most sellers put that on the listing information. I have identified weird patterns along the way and in order to make things easier for you, I am sharing them here.

Fujinon: The first number is the maximum focus distance in centimeters, the second number is the filter thread. For example Fujinon 16086 has its focus

limit at 160cm (which equals a +0.6 strength) and an 86mm thread. 190101 stands for 190cm maximum focus and 101mm threads. Fujinon has some great diopters, so keep an eye for them.

Canon: The newest series (250D and 500D) have their measurements in milimeters. 250mm stands for +4 and 500mm stands for +2. The “D” comes from “double element” which means they are achromatic doublets. The older series use similar measuring, like the 1300H, which equals 1300mm, 130cm or +0.75. The rare 900H has +1.1 power and features 105mm threads.

A good option to cover a wider range of thread sizes is the use of several step rings. The issue is that these can easily get stuck or lost. A different solution is to look for threadless diopters.

The best ones are called Series 9 and have a glass diameter of 83mm. With the glass at hand, you just need the right adapter. Taking the Series 9 as example, adapters were usually made by Tiffen and Kodak in sizes ranging between 67mm and 86mm. This means you can use the very same glass with lenses that have widely different thread sizes (67mm, 72mm, 77mm, 82mm and 86mm), covering almost every anamorphic currently available. Both adapters and diopters can be challenging to find, but when they come up they tend to be cheap. The adapters consist of two aluminum rings that screw on top of each other, holding the glass in between them. Then you just screw the whole thing in front of the lens just like a regular filter.



Adapters and rings for Series 9 filters.

There are lots of talk about achromatic diopters. What is so special about them? Their prices are way higher than regular diopters plus they are much harder to find. Fortunately they exist. These were first used in situations that require very short distances between lens and subject, and extraordinary image quality (the kind of thing you expect from a microscope, for example).

Single element diopters cannot deliver that level of quality across the field, creating strong chromatic aberration as you travel away from the center. In achromats or doublets, the two optical elements (a positive and a negative one) are combined in a way they cancel out their weaknesses and enhance the resulting image quality - especially around the edges. The holy grail of doublets is Tokina's 72mm +0.4, which goes for around US$350.

While looking for more information I ended up coming across a ***list of achromatic doublets*** made by various brands. It includes detailed information such as thread size, strength, rarity and even a price range (although very outdated). I also came up with my own shortlist of most common and desired close-ups along with their price range in 2013. Things have changed since then, but this is still good for reference.

- **Tokina +0.5 72mm – $150**
- **Kenko +0.5 72mm – $90**
- **Tokina +0.4 72mm Achromatic – $350**
- **Kenko +0.3 105mm – $350**
- **Canon 500D +2 72mm Achromatic – $100**
- **Sigma +1.6 62mm Achromatic – $20**
- **Angenieux +0.25 82mm – $330**
- **Kinoptik +1 82mm Achromatic – $530**
- **Foton-A +1 or +1.25 – $900**
- **Tiffen +0.5 to +2 138mm – $50 or more**
- **Tiffen +0.5 to +5 Series 9 – $1-50**
- **Tiffen/Kodak Series 9 Adapters – $20-40**
- **Tiffen +0.5 to +2 4.5" – $10-50 (rare)**

IIIG – ANAMORPHIC LENS-YCLOPEDIA

After a year of researching, buying and selling many lenses I realised how much time I wasted just looking for relevant and reliable information or tips about almost every lens. Some are so rare that it is hard to even find someone that owns another copy!

Based on this experience I decided to give something back to those who helped me along this path. I spearheaded the creation and development of a ***database*** with individual pages for each lens, including pictures, pros and cons as well as plenty of technical data such as weight, glass size, stretch factor, manufacturer, different names, custom clamps, attaching methods and so on.

The idea has gone far but it is challenging to find people willing to fill in the information for missing lenses. I am moving towards making it more acessible.

Currently the lens-yclopedia contains the most common lenses and some that went through my own hands or very close friends that volunteered to gather data.

The criteria to decide which fields to use were discussed at a ***specific thread*** created by me and relying on the input of several other anamorphic users at online forums.

IIIH – FAKING THE LOOK

In spite of all the magic involving anamorphic lenses, there are situations where it is much easier (and cheaper) to fake the anamorphic look rather than using actual scopes. Over the last couple years a bunch of different options came up to achieve the main anamorphic features: streak flares and oval bokeh. During research I had the opportunity to test some of these methods and evaluate their pros and cons.

*Author's note: If you want a much deeper guide on anamorfaking, try my new **Anamorfake It Until You Make It guide**, which packs 200 pages of original content, tutorials on modding several lenses, many post-production tutorials and techniques, and reviews of multiple products on the market, including everything that is mentioned in this chapter.*

Cinemorph Filter: Vid-Atlantic's filter comes in 58mm and 77mm sizes and has the goal of achieving both flares and bokeh through a very simple way. The Cinemorph is a UV filter on a rotating ring (like a CPL), in which good part of the glass is covered by a black vinyl, allowing light only through an oval area in the middle. Splitting the transparent area down the middle is a thin thread.



Cinemorph Filter – 77mm.

The black vinyl shapes bokeh. If light comes to the sensor through a specific shape, out of focus areas will display the same shape while everything else stays the same.

The nylon thread has the goal of "scattering" any direct light that comes upon it. Since it is very close to the front of the lens, it will always be out of focus, creating flares for strong and direct light sources.

The Cinemorph works well under ideal conditions but struggles in many situations. One of its struggles is staying still. The ring can rotate so you align the flares and bokeh as you wish but it does not feature a locking mechanism. Any bump or shake can spin it and make misaligned flares and bokeh. Its greatest advantage is keeping it simple, as easy as a spherical lens can be. The downside is due to all the black area, you lose two stops of light.

Based on an ***extensive series of tests*** using this filter combined with several focal lengths I learned it performs poorly with zoom lenses, casting heavy vignetting around the edges of the frame. On a full frame sensor it works well with primes from 85mm and up, which is a very restrictive number. Lenses wider than that end up capable of seeing the edges of the filter, ruining the trick (and the shot).

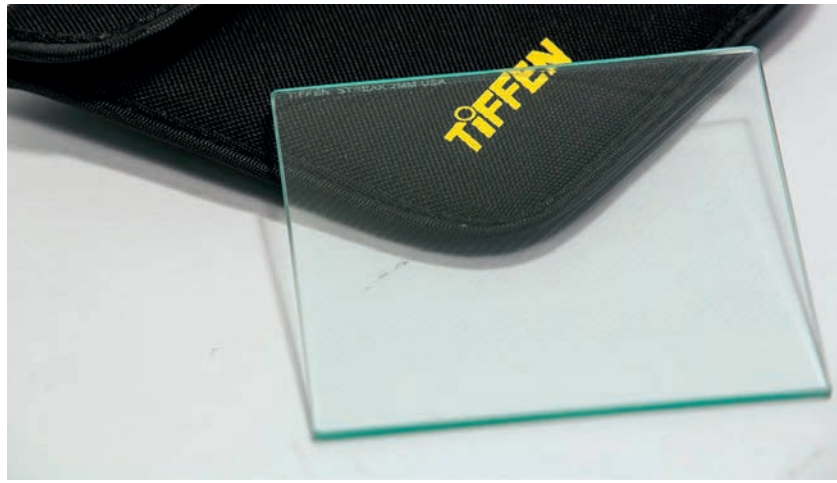
Besides that, all bokeh shows a thin semi-transparent across its center, cast by the nylon thread. Flares are super subtle.



Canon 5D Mark III + 85mm + Cinemorph filter.

Flare/Streak Filter: The Pro version of the Cinemorph filter. It does not try to achieve two things at once. It does exactly what the name says: it creates long

and pronounced flares originating from strong light sources. It is a 4x4 piece of glass with many thin lines that run across its width, all in the same direction. There are many variations for this filter, determined by two key factors.

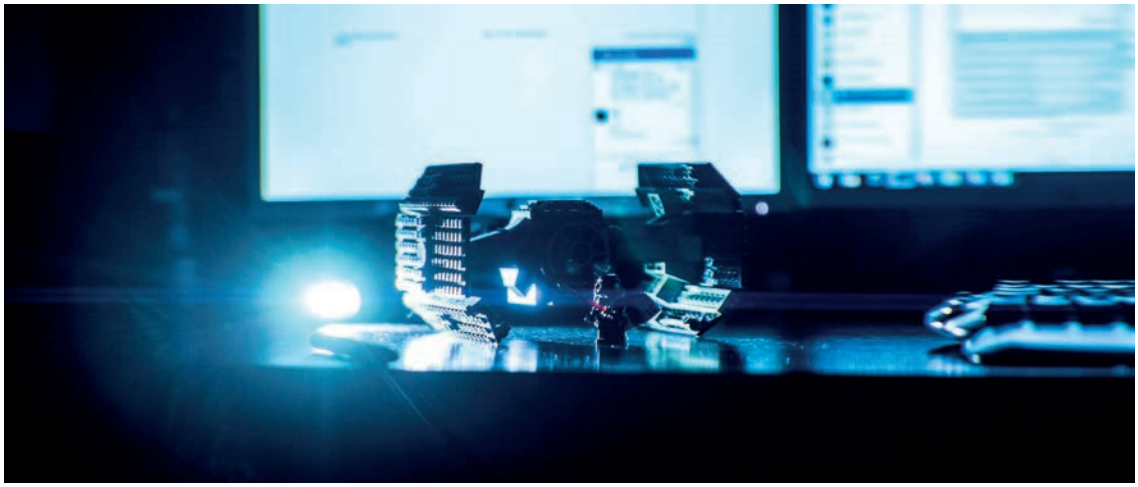


4×4 Tiffen – 2mm Streak Filter.

The first factor is color. With a neutral filter like this one on the picture above flares show the same color as the light source. If it is a bright blue light, flares will be blue. If you have multiple colors across the frame, flares will have different colors as well. There are other versions in which the flare is tinted to a specific color which can better trick your audience because most anamorphics do not change flare color, no matter the color of the light source.

The second factor is the spacing between the lines, measured in milimeters. The closer they are, the stronger the flares.

The following pictures are a comparison between real scope flares and the filter's. The first one was taken using an anamorphic adapter – Century Optics WS-13 – and the second one using the streak filter. Taking a closer look, the difference is easy to notice, but on a moving picture the effect is enough to convince.





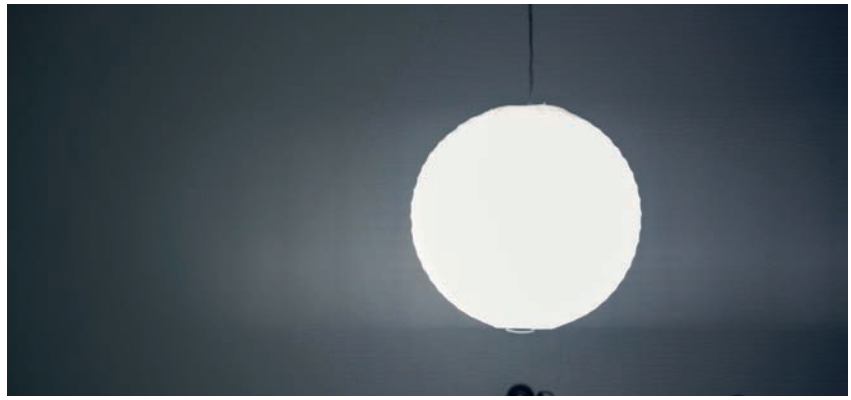
Comparison between anamorphic adapter and streak filter.

How can I disguise spherical glass even more? The following images were shot with a FlareFactory 58 lens. As the filter has 4×4 inches size, you are unable to screw it directly to the front of the lens and a mattebox solution is required. Unlike the Cinemorph, matteboxes have a proper locking mechanism for its filters so they will not rotate out of control.



As you can see in the picture above, the streak filter has side effects that resemble the Cinemorph's. Looking at the highlights on the left side, it is easy to see many thin vertical lines. These lines are the filter's lines which block light and affect bokeh.

Another occasion in which it fails miserably is when you have a large and soft light source inside the frame such as the paper lantern below. You can spot a large flare/smudge around the light with many thin lines – also caused by the filter. For smaller sources such as flashlights, LEDs, car headlights or non-diffuse lights, the filter is super effective.



FlareFactory 58: These lenses have among their strongest features good looking flares. Sold by ***Dog Schidt Optiks***, these are custom lenses, assembled from mods upon a prime lens, the Helios 44 (58mm f/2, M42 mount).

Among their customization options you can change contrast, aperture, external

paint and, more importantly, how flares look. These lenses make strong tinted flares. A quick look at the optics in the picture below and you will have an idea of Dog Schidt's madness. They are provided with Canon EF or Arri PL mounts.



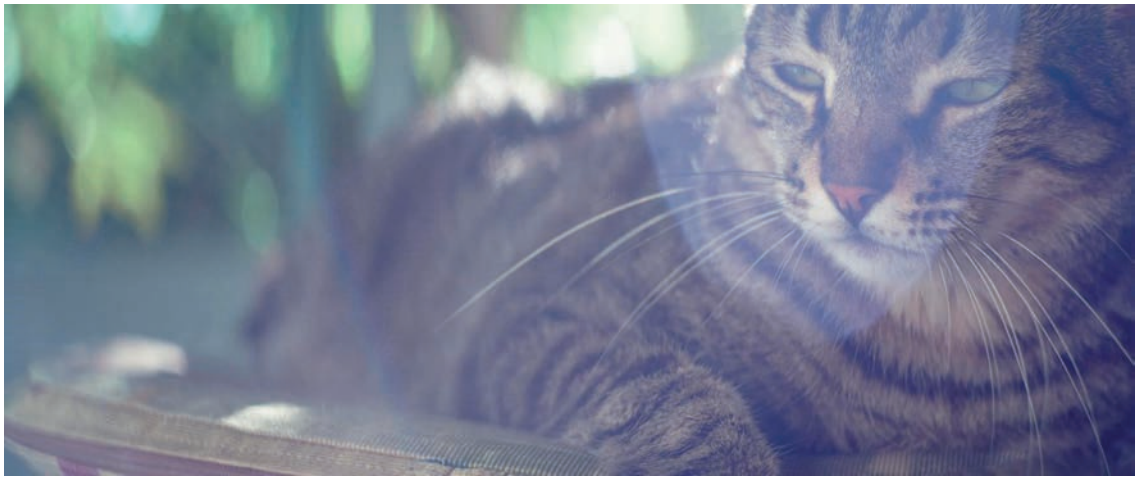
Image by Dog Schidt Optiks

But how do these colors affect only flares and not the whole image? It sure looks like the glass is tinted in there. After careful analysis and talking to Richard Gale – main head behind the project – I learned that what is tinted is actually the lens' inner walls between glass elements. These walls are usually black to prevent unwanted light bouncing and… flares. Now that they are tinted, the color comes through as light bounces around in there. There are ***many videos*** showcasing this feature.

Another mod they offer besides flare color is swapping the original aperture for a custom shape. DSO provides different shapes and sizes besides the traditional anamorphic oval (the downside is this mod sets the lens with a fixed aperture).

This idea was kicked off after Richard proposed, ***in a forum thread***, developing special lenses in order to enhance the results achieved with the most common fixed focus anamorphics (Panasonic, Century Optics and Optex). With these modified Helios 44 lenses it would be possible to achieve the oval bokeh aside from the tinted flares that make the look unique to each lens.

DSO lenses are expensive when compared to a standard Helios 44 lens, but when you take into account the huge number of mods they offer and how that makes every lens unique, price is not such a barrier anymore.



Test shot using a FlareFactory 58.

Richard has stated that longer lenses are easier to mod in order to make FlareFactories with different focal lengths while wide-angles are harder to alter. Other models are set to come out soon.

Modifying Lenses: The procedure is to open up a Helios 44, rip off the aperture blades and ***replace it with a laser-cut acrylic disc with a 2:1 oval shape in the middle equivalent to an f/4*** aperture. This way you avoid Cinemorph's greatest flaw, the black edges, and eliminate the need for an extra piece of gear.



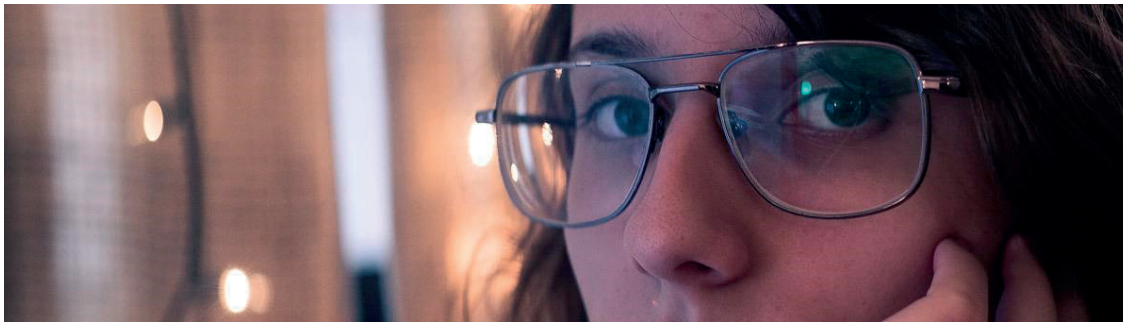
Modified 58mm f/2 Helios 44.

After replacing the aperture, its value is fixed at f/4, but it gives you 2x oval bokeh natively and allows for close focusing and rack focusing since you are still using a spherical lens.

I was very curious when I first came across this solution and decided to give it a shot. While the lens was in the mail I sent a “hi there!” message to Amir Safari – responsible for the video linked before – asking about the modification process and if he could share the file with the ***anamorphic aperture template***. A. Safari was a great guy to chat with, inspiring a lot of my experiments in future years.



Modified Pentacon Electric 29mm f/2.8.



Focus test using modified Helios 44.



Focus test using modified Pentacon Electric 29mm.

IV – MAGICLANTERN RAW

★ ★ ★ ★ ★

MagicLantern is an alternative software that runs on many Canon DSLR's firmware. A group of hackers and programmers started to study Canon's source code in 2009 and quickly achieved some advantages over the original firmware. From 2009 to 2013 the number of developers and users skyrocketed along with available – and stable – new functions.

How does it work? Following legal course, the code has been made from the ground up and is open for anyone willing to take a peek or implement changes. The developers avoid using Canon's functions because that is proprietary code – which could end up in piracy lawsuits. What developers are actually doing is tweaking the DIGIC processors that run the cameras, observing what happens and turning these experiments into custom features.

The software is free and can be download from its ***official website***. It is important to remember that even though the software goes through heavy testing, there is always a chance of coming across bugs or odd behavior from the camera due to MagicLantern's use. Canon's warranty does not cover gear damaged by third party software - although there are no records of any permanent damage so far.

MagicLantern's website displays a list with all compatible cameras and the features enabled on each of them. You can find the latest stable version for download as well as less stable versions including the day-to-day changes from various developers. One of the greatest achievements in this group is anyone can contribute: suggestions, tests, logs created by the cameras or even coded functions. Everything is read and replied quickly by an amazing team which does not get paid for this project.

By May 15th, 2013, MagicLantern's developers enabled raw video recording in Canon DSLRs. Not every camera can handle it or create decent results and much of the performance relies on fast memory cards. Still this is an amazing achievement, providing us with a feature that was only possible through professional grade gear, through expensive purchase or rent.

From the premiere of raw onwards I ran extensive experiments on short projects and stress tests to evaluate if this new feature was stable enough to go through a standard short-film production workflow. All the testing was rewarding: the quality boost is incomparable to the camera's original video codecs. The downside is we are eating through cards like there is no tomorrow.

Raw video's goal is to store the maximum amount of information within the frame, allowing for lots of manipulation during post-production. The

look straight out of the camera is flat and sometimes soft. From there the cinematographer can use all the light and color information stored on the clips to tweak and stylize them until the image reaches the desired final look. Raw recording allows for merging the latent image potential with the creativity of the person working on it.

The common ground between shooting raw with MagicLantern and using anamorphic lenses is the ability to pick a non-standard “recording window” from the camera’s sensor area. This way, when using 2x stretch lenses, I do not need to end up with a 3.56:1 (16:9 x 2) aspect ratio. If I set a recording window to something around 4:3 the final output is 2.66:1, slightly longer than the standard CinemaScope.





Shooting with a narrower aspect ratio than 16:9 you also avoid capturing the sides of the frame and then cropping them in post-production. This saves you precious file sizes.

IVA – CUSTOM CROPMARKS

My first use of MagicLantern combined with anamorphic lenses was designing custom cropmarks. These overlays allow previewing a different frame format than the one that is actually being shot. For example, I can have black bars covering top and bottom of the frame in order to make it look more cinematic, in 2.4:1 aspect ratio, while still shooting 16:9.

Mixing stretch factors in the same video was a common thing for me (1.33x stretch for some shots and 1.5x or 2x stretch for others). I knew something was gonna be cropped off later (either top-bottom or left-right of the frame). From there, I ***created nine unique cropmarks*** that allow mixing any kind of lenses (from spherical to 2x) highlighting what is the core framing and what will be cropped.



4:3 cropmarks for 2x stretch lenses

The file naming convention works like this: the first value is the final aspect ratio, according to the lens stretch. 1.33x stretch results in 2.4:1, 1.5x equals 2.66:1 and the monstrous 2x goes as wide as 3.56:1. The second part of the file name is for the kind of lens being used.

For example, my project was shot with 2x stretch and I am going for the 3.56:1 aspect ratio. After principal photography is done I have to shoot some extra footage using normal, spherical lenses. In this case I need to pick the “2x-norm” cropmark. This code can be confusing. The following image sequence aims to make it easier to visualize and understand the numbers.



Standard framing on the camera screen. It shows everything that is being shot.



1.5x cropmark on top of the image so you can compose your shot.



Final frame cropped during post-production.

In October of 2012 MagicLantern released an update for the Canon 5D Mark III, adding a series of new features. Among these was a fix for better previewing anamorphic lenses through an automatic squeeze of the LiveView image, displaying the corrected aspect ratio based on the lens stretch factor. This feature provides great results but eats a lot of processing power. Its only flaw is it does not work during playback. Developers informed me that messing around with Play Mode is quite complex and risky.



LiveView with ML's Anamorphic feature turned on.

V – ZONA SSP

★ ★ ★ ★ ★

Zona SSP started as a webseries project for a class in which I found space to experiment with everything we are not encouraged to try in traditional film production classes. I wrote and directed both episodes. I was also cinematographer, editor and responsible for all post-production (both sound and visual) in the project. My goal was not to pile up responsibility and create a one-man-band scenario, but since I was not able to find teammates to cover these roles in time (and budget), self sacrifice is what ended up happening.

The series' idea is to portrait a dystopic future in Brazil and make it look expensive. All this production value would be achieved using the simplest and cheapest tricks we could come up with without letting the audience know this was an extremely cheap film. At this stage I was saved by our super skilled art director, Luciana Parelho. Our lead actors, Lucas Durão and Larissa Orlow also jumped in without hesitation along with the other classmates. The full crew totaled less than ten people. We had lots of time for pre-production, which was key to get the script right, storyboards, visual references and to develop strong visual and sound concepts for the show.

Since the whole thing was a huge experiment I proposed we used anamorphic lenses in order to test them through production pacing. During those months we had student strikes taking place and the other graduation films I was supposed to work as director of photography looked like they were not happening.

Episode 01 is based entirely around daylight and exterior shots, relying exclusively on natural lighting and multiple locations. Episode 02 was shot almost entirely in a single room, at night, and has a much more tense feel to it. The lenses played a great part in how each episode gunned to achieve a more "cinematic" feel rather than TV or web.

Our budget was $150 for both episodes thanks to a lot of friends who borrowed us gear, wardrobe, accessories and valuable advice. I feel pretty proud of what we ended up with. All the editing and post-production including light reshoots and composing the score took us two months, which is also a very short turnaround for two 7-minute episodes.

To broaden the experience I wanted to use different kinds of lenses for each episode so I could better describe the process here. This change of lenses introduced a big change in overall gear, but we shall get to that later.

The main reason for changing lenses was the ability to compare the workflow in different scenarios - from the camera crew's responsibilities up to the format's specific needs. Plus I wanted to show how different lenses perform visually and have a final product to speak for me.

Since we had a low light environment and needed to work at wider apertures in Episode 02, I decided to go with the Original Iscorama, an Iscorama 54 and M42 Russian taking lenses. Episode 01 is day-based and has lots of bright sunlight, so I went with real cinema lenses, even though not as fast as the Iscoramas: OCT-19 mount LOMO zooms. Both systems will be described in detail over the following chapters.



M42 taking lenses and Iscoramas lined up for Episode 02

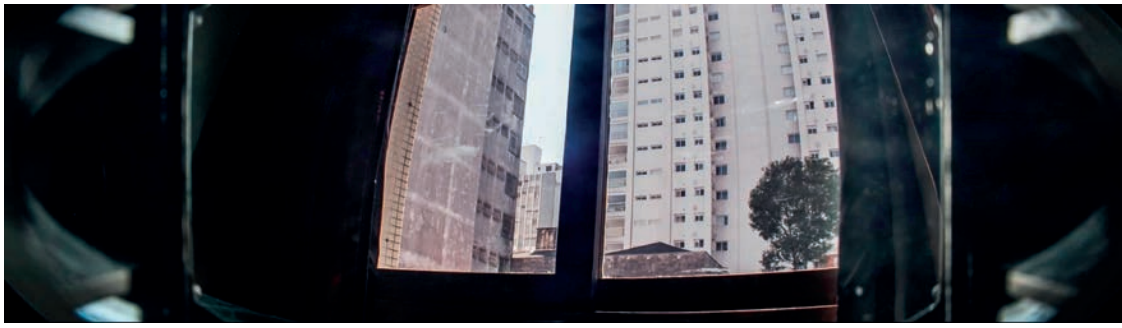
VA – LOMO – EPISODE 01

Episode 01 is a sunny chapter with lots of natural light, exterior shots and brightly-lit environments. The plan was to start early and shoot until sunset so we did not waste a single daylight second. Taking advantage of all this light we went with LOMOs, real cinema lenses. The ones I had required lots of light for optimal performance. LOMOs bring a low-contrast and unique look when compared to other optics I came across. When comparing the results, I think the first chapter is more cinematic than the second and a great deal of this feeling comes from the lenses' character.

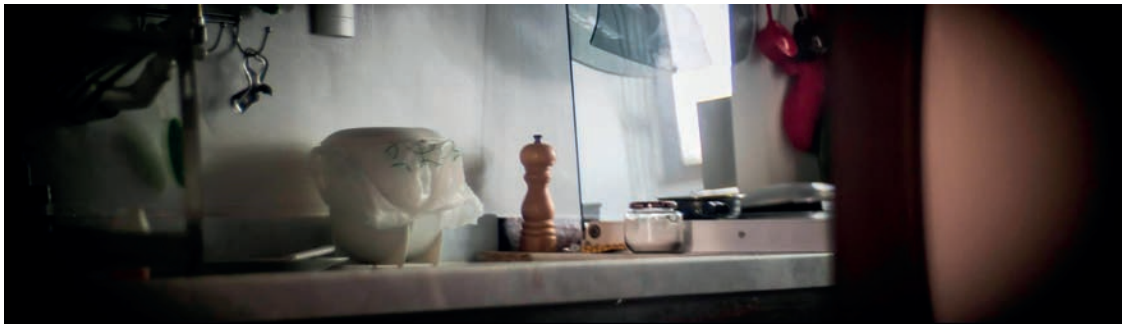
After the crew agreed to go anamorphic I started doing tests using an OCT-19 mount to EF adapter. The first problem came up where I was not expecting it: sensor coverage. Since these are cine lenses, they are designed for 35mm film. As the 5D Mark III has a full frame sensor – almost double the size of 35mm film – the lenses just created images on the central part of the sensor, leaving huge black edges around it.



OCT-19 to Canon EF adapter.



Canon 5D Mark III + LOMO Foton-A, @37mm (the inside of the lens shows up in the frame).



Canon 5D Mark III + LOMO Foton-A, @140mm (still shows heavy vignetting)

These black edges were way heavier than what I could handle in post, so I decided to swap the 5D MkIII for a camera with the proper sensor size. The chosen one was the Canon 50D, released in 2008. It was dirt cheap. Its sensor is almost the exact same size as 35mm film. Through MagicLantern and raw recording we set a 4:3 recording window that when stretched back to the correct values gave us 2.66:1 aspect ratio.

After picking the camera and solving the vignetting issue it was time to grab the lenses and go for real-world testing. Opposed to the various fast taking lenses used in Episode 02, on this episode I used only two big anamorphic zooms, manufactured twenty years apart.

The oldest one, made in 1969, is a LOMO Foton-A 37-140mm T/4.4. It is not a fast lens but its great zoom range allows for very distinct compositions. The anamorphic block goes in front of the spherical lens and focus is synchronized through bolts and screws that hold both blocks together.



LOMO Foton-A 37-140mm T/4.4

Minimum focus sits 1.6m/5.5ft away but we had a +1 diopter available, made specifically for this lens, which solved focusing issues in tight spaces and close up shots, increasing anamorphic bokeh as a bonus.

Buying, modding, servicing and shipping this lens could be a film on its own, going through Lithuania, Ukraine, Spain and finally Brazil over a span of five months between March and August of 2013.

First I needed to provide the seller a bunch of hand written documents assuring him I would be responsible for any problems with shipping the package to an address that was not my default billing address. Making it cross the border to Ukraine was a key part of the route for the lens to reach Olex Kalynychenko's very capable hands.

Once delivered, Olex swapped the original OCT-18 mount to OCT-19, declicked the aperture ring, installed standard focus and zoom gears, serviced both spherical and anamorphic blocks separately and then together. He also improved the connection between both blocks and developed a new support system for 15mm rails instead of the Konvas standard. After all this was done, my sister was in Spain and I asked Olex to ship her the lens. Finally, the Foton-A came home when my sister returned to Brazil.

The second lens, made in 1989, goes for the name of 350PF18-1 and covers the focal range between 20mm and 120mm with a T/3 aperture - considerably faster than its older sister. This lens uses a rear anamorphic adapter which kills every beautiful imperfection introduced by the bent glass (flares and bokeh, mostly), keeping only the stretch.



LOMO 350PF18-1 20-120mm T/3

OPF in Russian is an acronym for “lens with variable focal length” meaning this is a zoom lens. This 20-120mm in particular is an identical copy of a Cooke lens design but made in the Soviet Union.

One of its unique features is a macro ring embedded at the back of the lens, with strength from 1 to 5. 1 means standard operation without any macro capabilities. As the strength increases the maximum focusing distance becomes shorter and shorter, eliminating the need for diopters and allowing sharp focus less than one inch away from the front of the lens. I was able to focus on my fingerprint touching the glass.

VB – ISCORAMAS – EPISODE 02

For the second episode I went with the holy grail of anamorphic adapters, the Iscoramas. This decision was based on a key factor: their image quality and sharpness is excellent even at fast apertures like f/2 or f/2.8. Since my lighting scheme relies on small and dim light sources I must use fast lenses and the anamorphic adapter should not compromise my image quality.

In addition to the lenses, to impove image quality at low light we decided to use Canon's 5D Mark III which has great performance in our lighting scenario. This camera would also reduce the total amount of gear, as well as need less crew.

One of the greatest advantages of modern cameras is the fact that you can adapt many types of lenses flawlessly. For shooting Episode 02 we used a small set of primes composed by 37mm f/2.8, 58mm f/2, 85mm f/2 and 135mm f/2.8, all of them made by Zenit, the Russian optical company, around the 1960s.



M42 mount russian lenses and Canon EF adapter.

One of the strongest aspects of this set is the smoothness of their aperture ring, allowing changes during shots unnoticeably like a cinema lens does and unlike clicked stops from photography lenses. The fact that their optical design is so simple also improved the results of anamorphic pairing.

We had three kinds of Iscoramas for shooting. The first one was an Original Iscorama, modified to allow closer focus. We could go from 1.2m/4ft to infinity. The second was an Iscorama 54, much bigger and heavier, focusing from 2m/6.5ft to infinity, The third one, more of a backup lens, was an Isco Widescreen 2000 which has fixed focus, limiting our operation to diopter shots or far focus. We brought this one as a safety measure in case of accidents or risky shots and ended up not using it at all.



Iscorama 54

The issue of focusing subjects closer than 1.2m/4ft was solved through the use of diopters. We had a trio of achromatic doublets with powers of +0.4, +1.25 and +2, allowing us to change the Iscorama's focusing distance according to the framing needs.

As a good bit of the episode goes around a campfire we resorted to diopters in many shots to achieve sharp focus on our characters plus enhance flares and bokeh. The use of diopters was fundamental for all our close up shots.

The most important lens for this episode, the Original Iscorama, went through extensive upgrades before shooting. When I bought this lens it came in terrible shape with lots of scratches on the glass. I then got in touch with John Stelten from Focal Point Lens and sent him the lens for servicing.

Under John's watch, this Iscorama went through polishing and recoating of the front element which was the most damaged part, presenting a very negative impact on the resulting images. After it was fixed the Iscorama was sent back to Brazil. This process took me three months but the result was definitely worth it. Here are a couple of pictures from before and after servicing.



The lack of contrast and the huge halos around light sources were caused by damage to the front element.



After polishing, things went back to normal.



Just looking at the lens you can understand why the picture looked so bad.



After polishing only a faint scratch remained.

VI – THE PROCESS

★ ★ ★ ★ ★

This is the chapter where I will either regret the idea of shooting anamorphic or never go back to standard 16:9. It is a breakdown of the path I chose to follow during shooting and post-production. Its goal is to go beyond the theoretical – and historical – aspects of the lenses, diving into detail of how they behave in production and what kind of impact that had on the camera crew and the crew as a whole as well as the post-production process.

Changing aspect ratios is something that concerns not only the cinematographer. It also calls for decisions from the director and the art department. We took the CinemaScope chapter out of David Bordwell's "Poetics of Cinema" (2007) as our initial reference. The author goes in detail over the complications created by anamorphic lenses when they first appeared in the film industry and how after some time and lots of attempts Hollywood was able to embrace the new technology in a positive way. So positive that the CinemaScope aspect ratio is used even for films shot with spherical glass.

One of the good things about having a wider frame is being able to work better with negative space. For example, in Episode 02 negative space in the shots is quite dark, contributing to the feeling of not knowing what might be lurking around in that ruined world at night.

One of the smartest things I did to better understand the work was storyboarding on the ***proper aspect ratio***. This saved me from impossible compositions and physical issues when setting up the shots. It also got me familiar to the new frame size.

I will avoid going over things that happened in both episodes, writing about these aspects as a single event.

VIA – CONCEPT DEVELOPMENT

It takes a lot of imagination on the actors' behalf to put themselves into the context of this ruined environment that is not fully built around them. In this sense, shooting with a digital camera allows for smaller and lighter gear, decreasing distractions that might come from a large crew walking around or a noisy and large film camera. Besides, a lighter camera provides us with great mobility and versatility when setting up shots and capturing improvisation or coverage that was not planned on the call sheet.

We originally planned to use Canon's 5D Mark III but the plan had to change when the LOMOs could not cover the full frame sensor. We had to switch for an APS-C sensor camera. Canon's 50D was an unusual choice because even though it is a strong camera, it was already old, released in 2008, and if not for MagicLantern, it was not even supposed to record video.

The greatest feature about the 50D is that thanks to the enhancements provided by MagicLantern (better explained in ***IV – MagicLantern RAW***) this old camera was able to beat newer cameras in raw recording for it uses a Compact Flash cards instead of the more modern SD cards. Compact Flash can sustain faster data writing speeds while SD cards struggle.

Some of the visual and mood references we wanted to achieve through cinematography for Episode 01 come from the game The Last of Us (2013), and some of the jungle sequences from Hunger Games (2012). In both cases lighting is not stylized – the light is diffuse and does not show much contrast between highlight and shadow areas.

As for framing our sources were the movies The Book of Eli (2010), I Am Legend (2007) and some sequences from The Road (2009), which make great use of dynamic compositions between background and forefround in order to create tension.

A good example of this feeling is the first frame below, from The Book of Eli, where something takes place in the background but all the attention is focused on our character's face on the foreground. At the same time, the action and his face compete visually and it is hard to stay on his face for the whole shot, constantly going to background and coming back to check his reaction. They complete each other in terms of meaning and dismiss the need for cuts.






WICKED





From top to bottom: The Book of Eli, Hunger Games, I Am Legend and three frames from the game The Last of Us.

Since each episode was shot separately and our budget was a joke, we did not have proper (should I say “expected?) lighting gear - or even electricity running on set! All of our lighting was based on the Sun’s position in the sky, carefully picked times for each sequence and delicate handling of the raw files during post production.

The excess of natural light allowed us to work with smaller lens apertures, bringing more information to the background of each shot. This helped visually crafting the world we had in our minds: a city abandoned by humans and taken back by nature. We kept the camera set at lower ISOs since the aging 50D does not handle noise all that well - for reference, we were always below ISO640.

Episode 02 is the opposite. It takes place at night, it is dark and much more tense than the first episode. Because of this, cinematography required lots of contrast. For that we had numerous practical lights on set – torches, a real campfire, and strong LED flashlights representing moonlight. Since a lot of our lighting relied on real fire I worked closely with the art department to intensify or lessen the flames according to each shot, achieving a stylized look that is still realistic, with great shadow areas in every composition.

In order to keep the camera feeling from the first episode, we did not use a single tripod or any other support gear but the shoulder mount, giving me freedom to try long camera moves as well as creating a constant camera shake which works well for tension and increases the realism feeling we wanted for the series, coming closer to a documentary shooting style.

Episode 02 is strongly based on the evolving relationship between the characters. This begs for more close up shots and reactions than big camera movements or establishing shots. Going closer we increase the tension of what is around them at the same time we capture the essence of their performance and point of view.

Some of the references for framing, lighting and mood come from the movies Doomsday (2008), Hunger Games (2012) and more prominently The Road (2009).













From top to bottom: Doomsday, two frames from Hunger Games, and three frames from The Road.

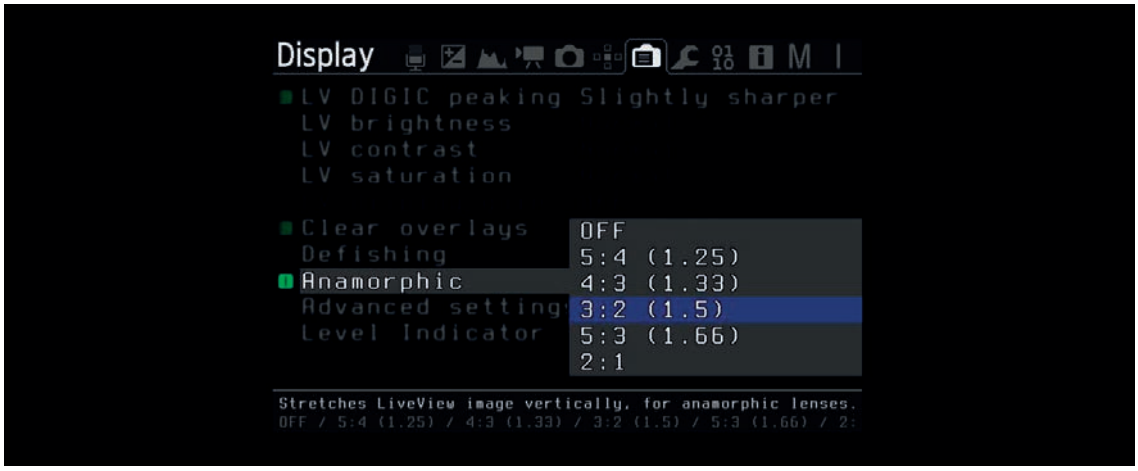
It is not a joyous and colorful look but it brings a great complimentary color scheme between the strong orange from the flames and the blue moonlight taking turns for the brightest areas in each shot. In Episode 02 we have no natural light at all! Its key lighting is the warm campfire and all other light sources are practical, allowing us to change framing quickly without moving light stands and other grip gear.

I have written about the lens choice for this episode on ***VB – Iscoramas - Episode 02*** and fortunately we did not have unpleasant surprises.

VIB – ON SET PREVIEW

One of my main concerns shooting anamorphic was how to look at the footage on set with the correct aspect ratio, a.k.a. unsqueezed. This is applied not only to when the camera is rolling but also during playback for evaluation.

Monitoring is easy to solve through MagicLantern's features. In the Display tab, there is an Anamorphic submenu. This submenu offers the most common lens stretches. By selecting one of the values the stretch is applied to the LiveView and the image shows unsqueezed while shooting.



Display Menu inside MagicLantern, Anamorphic option highlighted.

This option has two downsides. First the stretch is not applied on Play Mode. Everything looks good while you shoot, but not when you watch a previous shot in the card. The second downside is the LiveView distortion eats up a lot of processing power, which prevents us from shooting raw.

If you are shooting with the camera's native codecs or just photographs, this is a great solution that eliminates the need for extra gear devoted only to fixing the images' aspect ratio.

Our case was not “Tito shooting and walking around”. We had select locations, call sheets, storyboards, so a larger number of people needed to see the shots as they played out (director, art department, sound, and so on, each department looking for specific bits of information whithin the frame related to their work). This multi-department need calls for a screen bigger than the camera’s.

Unfortunately not many external monitors feature anamorphic stretching among their menus. During pre-production I looked for some of them based on ***tips from other anamorphic users*** and specs from the manufacturers’ websites (only ***Zacuto***, ***SmallHD*** and ***Cineroid*** provided this feature when this was written). Some ***TVLogic*** models present a feature called User Aspect in which you can set specific dimensions for the height and width of the frame, similarly to a user-made stretch.

I already owned a Zacuto EVF which covers the most common stretches – 1.33x, 1.5x and 2x. Image quality is excellent but the screen size is exactly the same as the camera, only 3.2”. Its performance is improved through the use of a loupe which allows you to pay close attention to every bit of the frame.

The Cineroid EVF4 has a 4” screen and it is not much bigger than the Zacuto EVF. It also has a loupe that must be used along with the small screen for ideal operation. I could not test it in person, but the manual states that you can apply any stretch value over the displayed image through custom parameters.

Saving the best for last, SmallHD offers a wide range of monitors with anamorphic stretch. The DP4, DP6 DP7 and AC7 models are capable of anamorphic stretching for 1.33x, 1.5x and 2x lenses. The model number is equal to its screen size in inches so none of them is really huge - the biggest one is 7”.

A common feature to all these monitors and EVFs is they provide “in” and “out” ports for HDMI and/or SDI connections while cheaper models have only an “in” port. Thanks to this pair we can chain them one after the other to make the camera’s image easier to share around the crew. In our case we used the Zacuto EVF paired with a SmallHD AC7. Since the Zacuto is small and light, it was stuck to the camera rig as my viewfinder while the bigger screen was further apart for the other crew members to use.

That would have been enough if we were not shooting raw. Reviewing ML raw material straight from the camera is still pretty terrible. It only plays a couple frames per second and does not show any color - this pretty much stands for “unusable”.

The solution we found was to use an external recorder – Atomos NINJA 2 – with HDMI input as the first link in our monitor chain. Its recording was triggered directly by the camera and it was easy to review the footage. From the NINJA the image stream would go through the Zacuto EVF which stretched the frame and then passed it on to the SmallHD AC7, also with corrective stretch.

With this number of small bits of gear and several meters of HDMI cables we were able to see each shot in the correct aspect ratio as well as review them using the external recorder to decide if we should move on to the next shot or do another take. The additional monitor also allowed the crew to stay further back instead of hovering over the cinematographer, which sped up the work.

VIC – POST PRODUCTION WORKFLOW

Author's Note: The math in this chapter has been updated to current 4K standards and it is way more elaborate and accurate than the original content of this chapter - which was aimed heavily at MagicLantern raw processing. You can still find the original VIC chapter in the Appendix at the end of this guide.

Back in the film days, if a motion picture was shot using anamorphic lenses, after the cutting process, the same image frames would be projected back onto the screen. The film print would still be squeezed. In order to reverse the effect projectors were also fitted with anamorphic lenses.

If you have an anamorphic lens at hand, notice that while you look down the barrel from the front of the lens the image gets squeezed, but if you look from the back, the image gets stretched!

It is because of this solution that we ended up with so many of what are called "projection" anamorphic lenses. Many of the adapters used today were originally used in theaters. It was not until digital really took hold that folks needed to desqueeze the image for post-production - editing and visual effects.

Alhtough digital post-production normalized the squeeze of the picture, the final products were still squeezed back on the celluloid rolls that went to theaters, to be projected by an anamorphic lens.

The easiest solution that digital intermediate found for not altering the image in irreversible way was to ***assign a non-uniform ratio to every pixel in the frame***. Pixels are usually square, but once you tell the computer that each pixel should be 33% wider, the resulting picture is 33% wider. This is just an interpreting process and it does not change the values of the pixels. A 4K film scan with 1.33 pixel aspect ratio does not "become" a 5.2K film scan.

Things have changed a lot since digital was first introduced into film and now there is no celluloid at all - at least for you and me. Our cameras shoot digital pictures that can only be read in a computer and we do not print the final results

back onto film rolls, but upload them online. The next few pages are going to cover popular aspect ratios, combinations of shooting aspect ratio, anamorphic squeeze and final aspect ratio, and lastly you have video tutorials for Adobe Premiere, Photoshop, After Effects, DaVinci Resolve, and Final Cut Pro X.

Up until 1973, the famous CinemaScope aspect ratio was 2.35:1. This means that the width of the screen is 2.35 times bigger than its height. Then, after 1973 the optical soundtrack was removed from the film print and that widened CinemaScope to 2.39:1. As a rule of thumb, 2.4:1 is close enough to be accepted as CinemaScope as well.

Some of the most common and desired ratios are featured in this cheat sheet. You can set any two variables and find the third value.

Shooting Frame	Lens Stretch	Final Aspect Ratio
16:9	1.33x	2.35:1
16:9	1.5x	2.66:1
16:9	2x	3.56:1
4:3	1.33x	16:9
4:3	1.5x	2:1
4:3	2x	2.66:1

As you can see, none of the combinations yield perfect 2.4:1 aspect ratio. On a 16:9 sensor that would require a 1.35x scope and a 4:3 sensor requires 1.8x stretch. If we flip the chart on its head, we can find our shooting aspect ratio.

Final Aspect Ratio	Lens Stretch	Shooting Frame
2.4:1 (CinemaScope)	1.33x	1.8:1
2.4:1 (CinemaScope)	1.5x	1.6:1
2.4:1 (CinemaScope)	2x	1.2:1

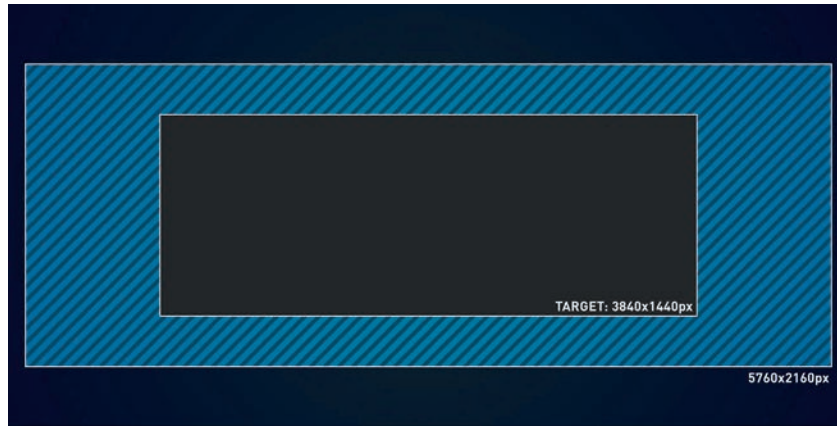
It is always better to overshoot than to end up with too little. The last step is discarding this extra side area in the editing/post-production stage. For example in Episode 01 we used 2x stretch lenses. We should shoot with a 1.2:1 proportion to end up with a 2.4:1 aspect ratio . The closest to 1.2:1 that

our camera could do was 4:3 – which reads as 1.33:1 – just a bit wider than what we wanted and delivering 2.66:1.

This all goes to say you should know your final aspect ratio before you start shooting your project. Your final aspect ratio will determine the combination of lenses and shooting aspect ratio to be used during production. At least one of these parameters is really hard to change. Some cameras do not offer 4:3 shooting. Or you might only have 1.5x lenses. Pick the format that will give you the least amount of wasted footage based on the reference chart above!

Once the shoot is done, it is time to bring all the footage into your editing software and figure out your final frame sizes. In today's world 4K is king. You can shoot higher resolutions but very rarely someone will ask for a delivery larger than 4K. The vast majority of us is not even equipped to watch 4K content properly. Unless you have very good reason, I would avoid setting resolutions higher than 4K or C4K.

Something to keep in mind is your final shot size is unlikely to match your timeline resolution perfectly. For example, if you want your project to end up in CinemaScope 2.4:1 but you are shooting 4:3 with a 2x stretch lens, this is what happens:



There is a little bit of overflow on both sides. That can happen on the top and bottom of the frame as well, depending on shooting aspect ratio and squeeze factor. It is normal and, contrary to popular belief, you are not losing a ton of resolution.

It is time to look into how to rescale your footage and set your timeline resolution! The timeline settings are easier, so we will start with those.

First, pick your width. 4K? 2K? Full HD? Something else entirely? It is up to you. For the following examples I am choosing C4K (4096px) and 4K (3840px). Next, pick your aspect ratio. 2.4:1 CinemaScope? 2.2:1? Again, it is your choice and how it serves your project. I will do two scenarios, one for CinemaScope and one for 2.66:1 because I use this aspect ratio more often than I expected.

The next step is to divide your frame width by the aspect ratio.

So I will divide C4K (4096px) by 2.4.

4096 ÷ 2.4 = 1706.6666…

Round this number to the closest even integer. In this case, 1706.
Our timeline resolution in this project will be 4096x1706px.

Again, but now for 4K (3840px) and 2.66:1

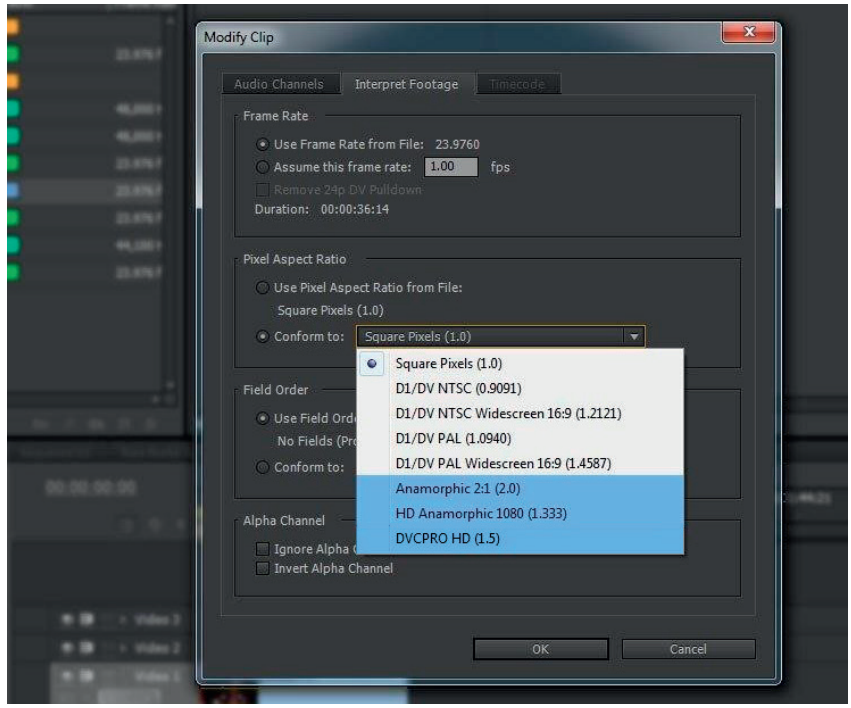
3840 ÷ 2.66 = 1443.60

Which I will round to 1444 and set my timeline resolution to 3840x1444px

In an ideal world you also make the timeline height a value that gives you an integer when divided by 8. This is not mandatory in terms of aspect ratio but it will improve the computer processes and give you more subpixel accurate results.

Taking this into account, 1706 would become 1704 and 1444 becomes 1440. I prefer to round them down.

Once your timeline settings are out of the way, it is time to rescale your shots. The first approach is to simply multiply their width by the lens' stretch factor. The easiest way to do that is to change the pixel aspect ratio for all the shots. This is usually done before dropping them on the timeline. The most common anamorphic squeezes are supported in pixel aspect ratio menus (1.33, 1.5 and 2x).



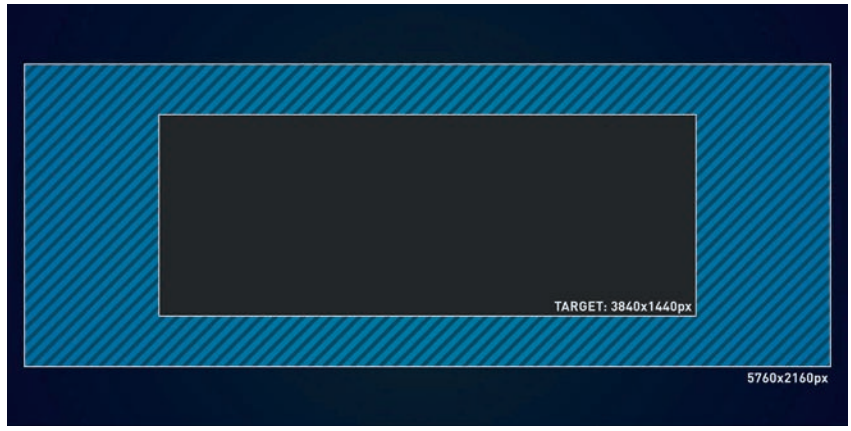
Pixel aspect ratio options available in Adobe Premiere

As an alternative to using pixel aspect ratio you can manually rescale each shot once they are in the timeline. There are two different methods for this: one involves changing the width of the frame, the other one changes the height. A third method, more hardcore, is a combination of the previous two, changing both height and width values for an optimized result. The important bit for all of these methods is that the ratio between width and height values matches the lens' squeeze. For a 2x lens, width scale divided by height should always be 2, regardless of how much you change them.

For the first method, changing the width, all you do is multiply the width scale by the lens' stretch factor. For example if you are using a 1.5x lens your scale parameters are 150% width by 100% height, or 1.5 by 1.0, depending on how the software takes them.

This method easily makes your shots too big though. For example if your timeline settings are 3840x1440px as shown above and you are shooting 4K on a 16:9 sensor using a 1.5x lens, this will give you:

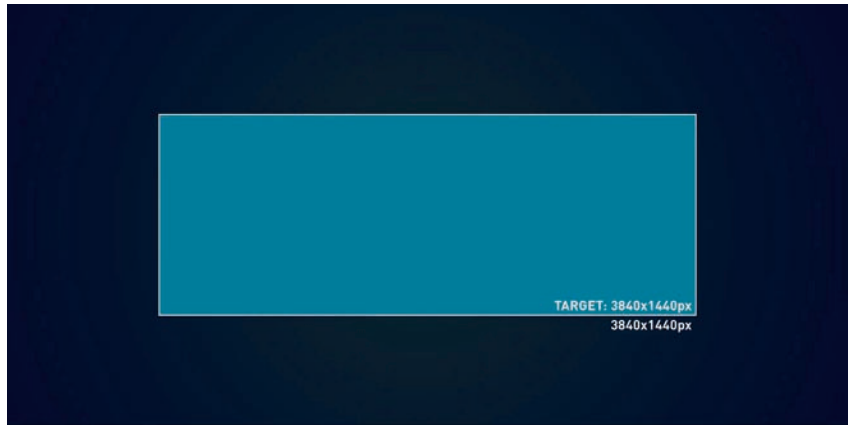
*(3840 * 1.5) = 5760px width by 2160px height.*



Both numbers are wildly off when compared to the target (3840x1440) and we are creating a resolution much larger than 4K. So we explore the second method: rescaling the height. In this process you will divide the shot's height by the lens stretch factor. In terms of percentages, this looks like 100% for width and 66.67% for height, or 1.0 and 0.67.

Using the same example, shooting 4K on a 16:9 sensor using a 1.5x lens, this will give you:

3840px width by (2160/1.5) = 1440px height.



A miracle! Perfect match! This second method also preserves more detail since you are scaling things down instead of up. This perfect match usually does not happen in real life though, which leads to the third method: combining a scale up in one dimension and scale down in the other, but keeping in mind that dividing width by height gives us the lens' squeeze.

For this scenario I will use the GH5 shooting 4K in 4:3 aspect ratio (3328x2496px), using a 1.5x lens. Same timeline resolution: 3840x1440px.

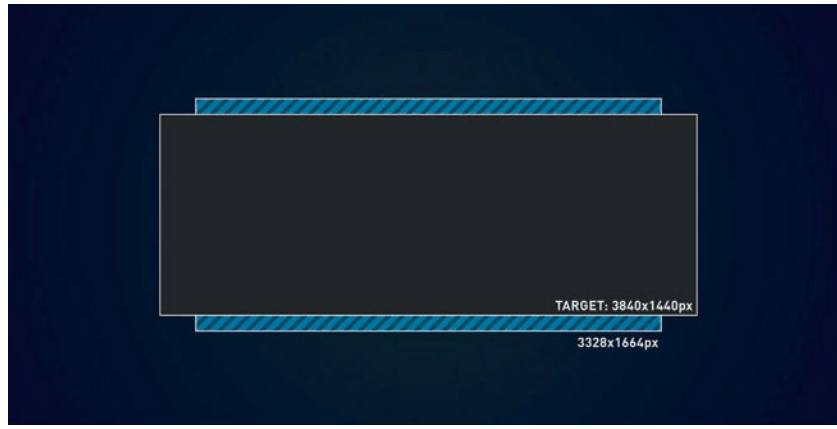
If we stretch the width:

*(3328 * 1.5) = 4992px width by 2496px height.*



We are off on both values. So we try method number two, dividing the height:

3328px width by (2496/1.5) = 1664px height.



Still off. So method number three: we divide the target pixel size by our current pixel size.

3840 / 3328 = 1.15

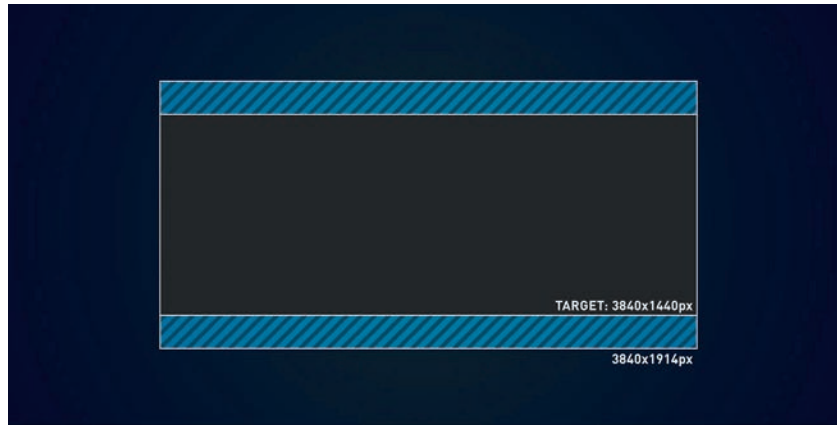
1.15 (or 115%) is going to be our width rescale factor.
Now we divide that by the lens' squeeze factor (1.5x)

1.15 / 1.5 = 0.76666

0.7666 (or 76.66%) is going to be our height rescale factor.
Now we put all of that together:

3328 * 1.15 = 3840px

2496 * 0.7666 = 1914px



This gives us a 3840x1914px final size. It is still not a perfect match to our target (3840x1440) but I warned you about this in the beginning. The spare top areas are going to be cropped out or used for reframing!

In the Extras folder of this guide you will find video tutorials for many different software if you want practical examples beyond the pure math presented here, plus a handy calculator to skip all this math for your timeline settings and scale ratios.

VII – CONCLUSION

★ ★ ★ ★ ★

Looking through the lenses is a must in order to understand how they work. The time I spent researching and reading online was elightening but it does not come close to the very first minutes handling each lens and seeing the world through it. The ideas and concepts are quite confusing until you get to the practical side of shooting. That is when articles and tests fit together and the use of anamorphics makes perfect sense. Having the lens in your hands does not just illustrate what was read before, it is much more than that. Shooting with scopes creates new questions and drives more research.

When I mounted my first lens to the camera I struggled before getting anything in sharp focus. A couple more months passed until I got used to aligning scopes quickly and found proper diopters.

Did I ever think about quitting and going back to what I used to shoot before? Yes. There were times I felt the gear refused to work and the anamorphic picture remained in hiding. Whenever that happened I did not bat an eye before unscrewing the adapter out of my taking lens and shooting spherical. After each of these crashes I went back a couple steps, re-tested combinations, figured out the ways of each adapter and worked on the problems that made set experience harder.

It was just recently that I felt safe enough to decide whether a lens is just a “good lens” or if it is a “lens that works well with my shooting style”. This was only possible because I took plenty of time to play extensively and carefully with each and every one of the anamorphics that I came across.

The anamorphic format and personality are so strong that they pushed me past the point of no return. I got rid of almost all my spherical lenses, keeping only the gear that fits an anamorphic workflow. I feel I was “photographically reborn”, leaving behind practices I knew as standard and safe. From there I explored new and more demanding techniques, seeking visual excellence and a different feel from what I saw on Brazilian cinema, TV or web productions.

The most amazing thing about shooting anamorphic is the community of people passionate about it. So much passion makes it quick and easy to learn, experiment and get feedback. The welcoming community was a key feature to keep me on track for my own tests. Along the process I exchanged words with folks from all over the world either asking questions or providing answers. During this journey I made many friends in conversations that went further than mere advice on buying/selling lenses or specific questions. I am still close with most of the people I met during this process. Having an influence from multiple different cultures enriches the experience and makes you think about perspectives you otherwise would not.

Through this guide I aim to encourage experimenting, research and the use of anamorphic lenses in smaller productions. There are plenty of curious people out there. If my work is enough to provide them with a good starting point, answering some of the most frequent questions, you can then move on to your own pace into the subject. Inspiring others to go down the same path I did is plenty to keep me energized. This is not a definitive guide.

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★ ★ ★ ★ ★

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IX – WHAT LIES AHEAD

★ ★ ★ ★ ★

It has been a year since the guide was originally translated. During this year, a lot happened. SLR Magic's Anamorphot came out, we got the ***Focus Module***, 4K is becoming very popular, the GH4 allows people to shoot 4:3, ***Rectilux*** is just around the corner, and so on.

I tried to follow how things are going, but there is nothing like testing in person, and my budget is currently non-existing. I moved from Brazil to Canada after a career in VFX but cinematography and anamorphics are not forgotten. Over the following months I have some lenses to test and review. I placed an order for a Rectilux and got a Kowa B&H from a friend. I got an Isco 16:9 Video Attachment, I finally got an Iscorama 42 and most definitely will try something for the ***FM lens contest***, so these might have their posts here soon.

There are also more specific products that I would like to try, out of curiosity, like ***Moondog Labs' anamorphic adapter for iPhone***, the tiny ***VM lens*** paired with something small on the BlackMagic Pocket Cinema Camera, the new ***Cinemorph filter, that goes on the back of the Sigma ART 18-35mm f/1.8 lens***, and the – not so new anymore – ***SLR Magic's Anamorphot*** and ***Letus AnamorphX***. I have seen test videos and read reviews, but they are not as deep as I want them to be. I know what to expect from each lens but the exact details and feeling are not there. Test videos are nice but you cannot figure out how that would work in a real production workflow, what quirks they have and how to benefit or get around them.

I was able to find the SLR Anamorphot on a rental house nearby and might try going there during school's break to test it out, but most of the other lenses are completely out of my reach and I am not willing to spend any money to make a review and deal with the trouble of passing the lenses forward… We will see what happens.

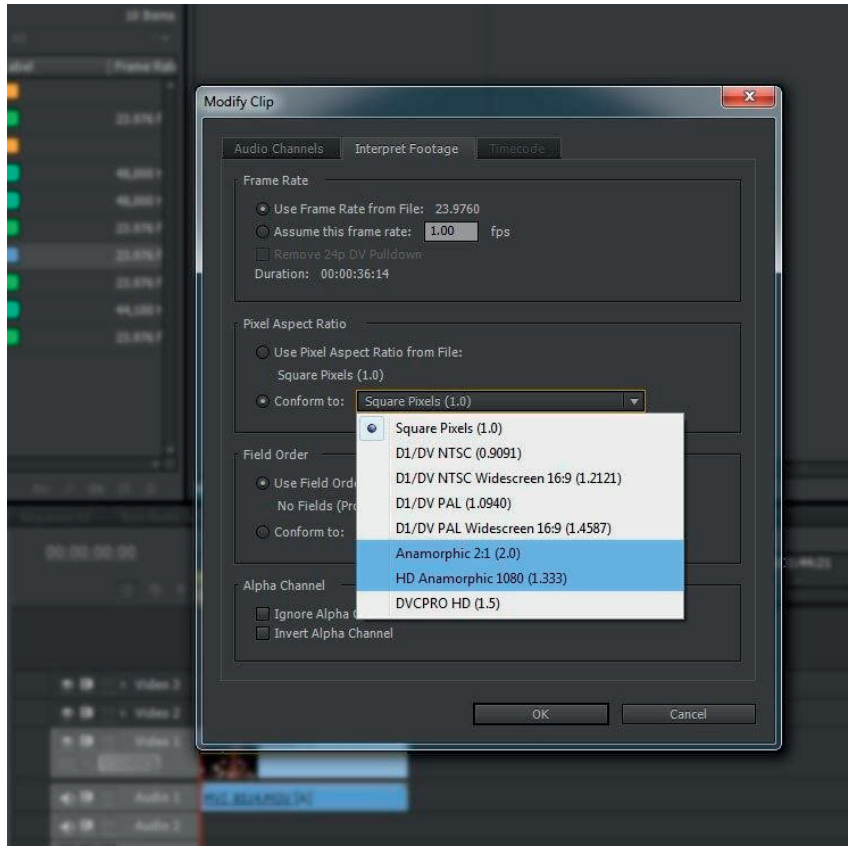
APPENDIX – ORIGINAL VIC – WORKFLOW

★ ★ ★ ★ ★

DISCLAIMER: This was written long before the MLV format and real time players for MagicLantern's raw files were released! More on that in the future, I hope.

The most complicated step on this process is dealing with the raw footage. The anamorphic bit is simple to work out, but both aspects are explained on this chapter starting with anamorphic and not taking raw into account at first.

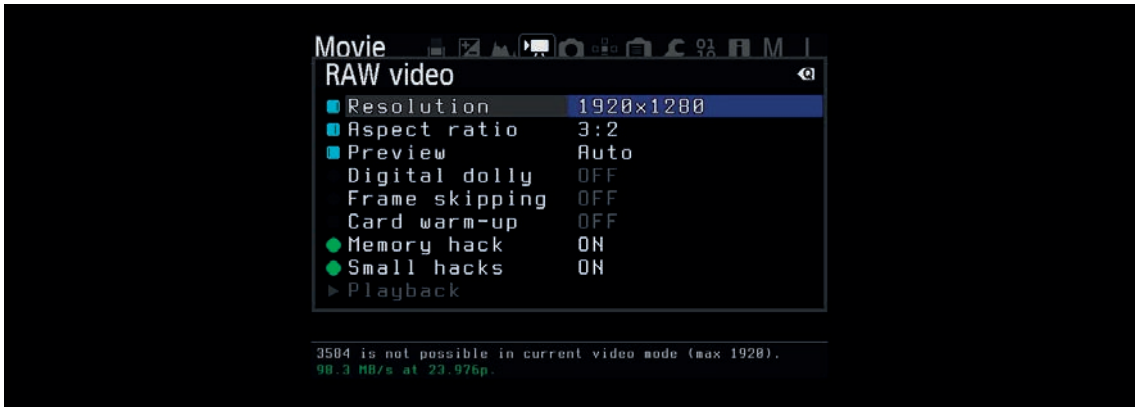
If the footage was shot on the camera's standard codec we can stretch it out to the correct aspect ratio directly in the editing software (FinalCut Pro, Adobe Premiere, Sony Vegas or whatever software you use) by changing the clips' ***pixel aspect ratio*** according to the lens stretch factor.



Pixel aspect ratio options available in Adobe Premiere CS6

Working with anamorphic still images requires two steps, though. The first one, importing the raw image into Adobe Lightroom or Adobe CameraRaw (or any other raw processing software of your choice) and adjusting exposure, contrast, highlights, shadows,etc so the image looks like what you want. The second step is loading it on Adobe Photoshop in order to fix the stretch. Most raw processors do not offer support to pixel aspect ratio changes or variations in the width/height of the images.

Now raw: the first step is to set the camera to shoot with the proper resolution, taking into account the lens' stretch factor. Inside MagicLantern's menu, this is easy to set up.



MagicLantern's RAW Video Menu

Some of the most common and desired ratios are shown below as a quick cheat sheet where you can set any two variables and get the third one.

Shooting Frame	Lens Stretch	Final Aspect Ratio
16:9	1.33x	2.35:1
16:9	1.5x	2.66:1
16:9	2x	3.56:1
4:3	1.33x	16:9
4:3	1.5x	2:1
4:3	2x	2.66:1

Final Aspect Ratio	Lens Stretch	Shooting Frame
2.4:1 (CinemaScope)	1.33x	1.8:1
2.4:1 (CinemaScope)	1.5x	1.6:1
2.4:1 (CinemaScope)	2x	1.2:1

The aspect ratios on the second table (1.8:1, 1.6:1 and 1.2:1) are not perfectly achievable in MagicLantern's menu so the way to go is pick the closest proportion to what you want but a little larger than your desired Final Aspect Ratio. For example in Episode 01 we used 2x stretch lenses so we should shoot with a 1.2:1 proportion which is not available on the menu. The closest to 1.2:1 available is 4:3 – which reads as 1.333 – just a bit larger than what we wanted. It is always better to overshoot than to end up with too little. The last step is discarding this extra side area in the editing/post-production stage.

Due to these different window sizes, files end up with different sizes as well and each card may store a little more (or less) footage. For reference: while shooting 16:9 in Episode 02, 64GB cards allowed us to shoot ten minutes of footage. For Episode 01, since the window was smaller (4:3 compared to 16:9) each card lasted 17 minutes which is a considerable difference in the pace of card switches.

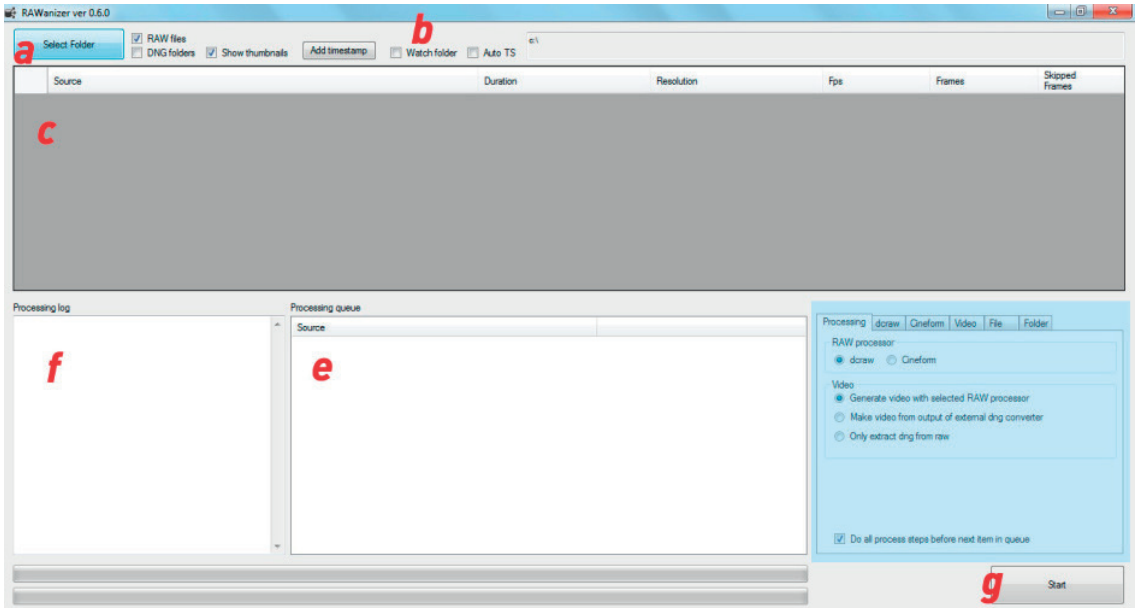
After setting up the camera and shooting until a card is full we get to the thickest part of the workflow. There are ***several options for these steps***, but here is the one I used in both episodes. My goal was to keep it as simple and straightforward as possible.

The card goes into a USB 3.0 reader and the files are copied over to the movie's hard drive – also USB 3.0 since the files are massive and we have to go through them as quickly as possible to free the card so it can go back to the camera crew.

The files are not "playable" at this point, they are simply containers that store the DNG frames (until now there is only one software that can play them natively, the ***Drastic Preview Professional Media Player***). In order to keep going down the flow we must extract these frames and generate proxy versions (lighter files with a lighter encoding such as Apple ProRes) for each clip. These proxies will be used for the edit and later replaced by the DNG sequences which give us a LOT of power for color correction, adjustments, effects and post-processing. They are VERY heavy, though.

For this extraction process we used ***RAWanizer***, a software developed by a MagicLantern user fulfilling the requests made by several forum members. There are plenty of software options available for this extraction and conversion process - like ***RAWMagic*** or ***raw2dng*** - but for our Windows machines RAWanizer was simple and quick to use.

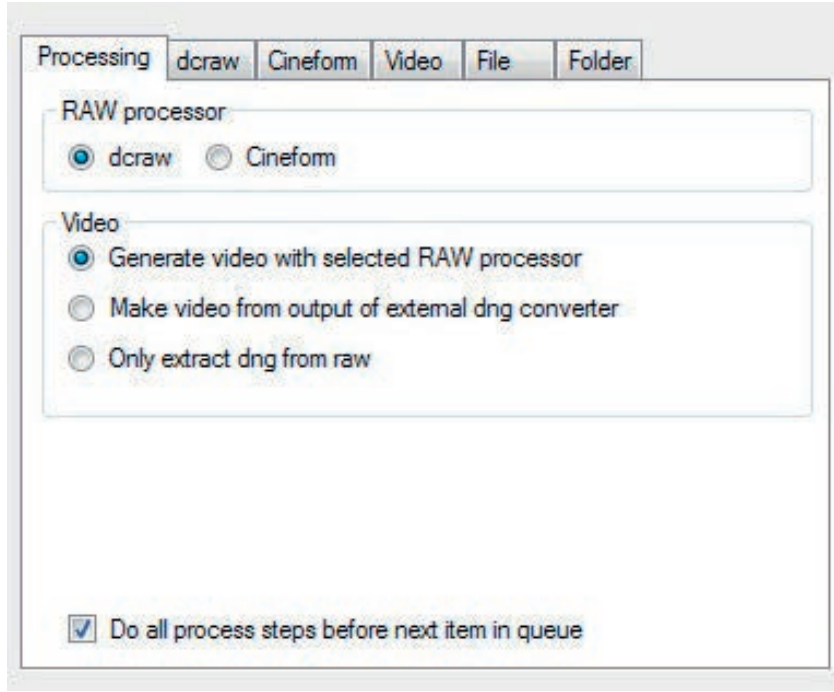
Once you have it installed, follow the steps below for processing:



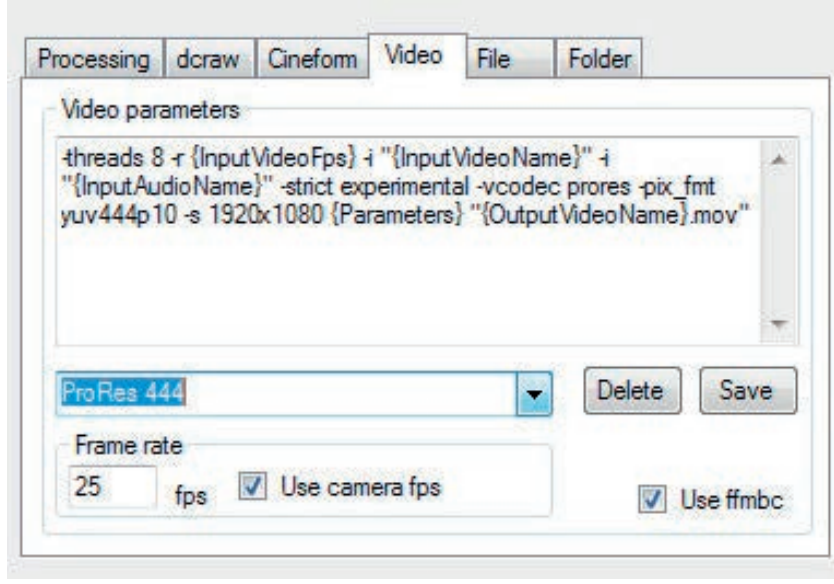
Click on Select Folder (a) and navigate to the folder where the raw files are. Depending on how you are loading the files and organizing them, it might be a good idea to mark the box for Watch folder (b) so the software keeps looking for new files recently copied from cards to the same folder. Be careful when checking this option if you delete the watched folder as the program might crash while trying to find it.

After selecting the folder, all the shots inside it will be loaded into the grey area (c). If you check the Show thumbnails box RAWanizer will grab sample frames from the beginning, middle and end of each shot for preview.

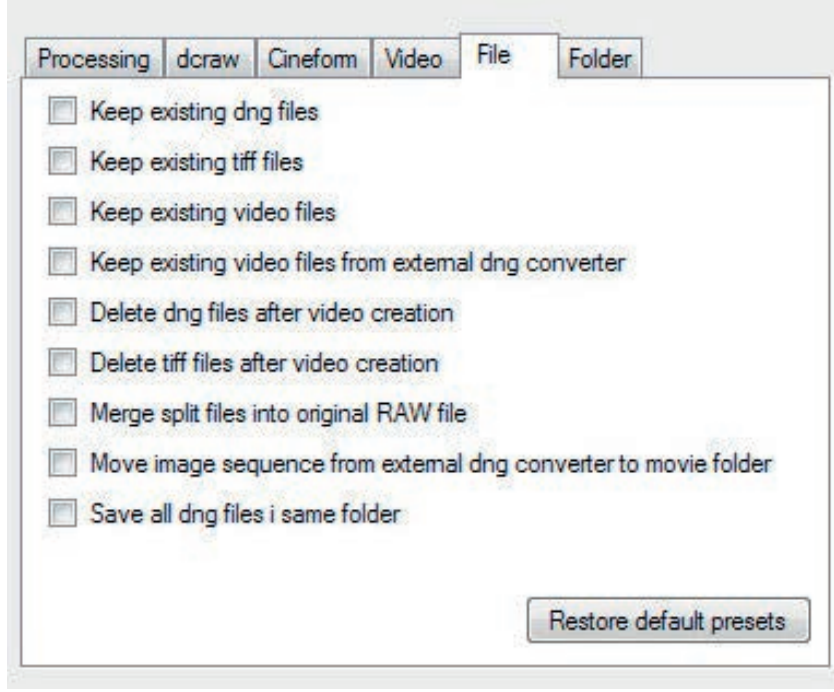
The following steps will happen in the highlighted blue square.



On the Processing tab you set the steps the program will follow: if it just extracts the DNGs from the raw container, if the processing is to be done by an external converter or if all steps (DNG extraction and proxy rendering) must be completed for each clip before moving on to the next one.



In the Video tab, you choose the codec that will be used for the proxies from the dropdown menu (the one that says ProRes 444 on the image). It is also important to specify the video's frame rate even though some times this information comes straight from the camera. With a little more advanced knowledge you can change the codec's settings through code and even stretch the proxies to the correct aspect ratio at this step.



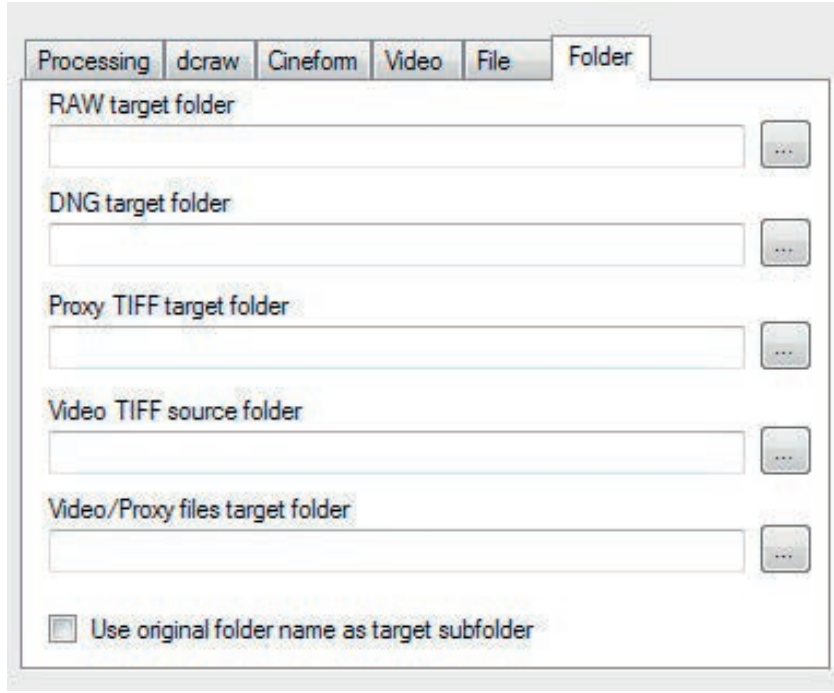
In the File tab we check the boxes that must be executed after the raw file is processed. The ones that worked for us were "Keep existing dng files" because even if we tried to process the same file twice, RAWanizer would detect the duplicate files and skip them altogether. The same goes for "Keep existing video files" for avoiding double proxies.

Then the options for "Delete dng files after video creation" would erase the files we want so much for post so no checking that one. "Delete tiff files after video creation" was useful since these files were not being used for anything else down the line.

Since many cards were being formatted inside the camera and it follows FAT32 file system, shots bigger than 4GB were split in more than one container. The option "Merge split files into original RAW file" forces RAWanizer to stitch these containers back together before processing the shot. If this box is not checked the software will just go through the first 4GB of the clip and skip the rest.

A way to avoid split files altogether is formatting the cards on the computer and selecting exFAT file system, which does not limit file size.

The steps taken by RAWanizer are: first, check if the shot has more than one 4GB container, if yes they will be stitched together. Then all the DNG frames contained inside the raw file will be extracted. The slowest step comes next when generating the proxies: every DNG frame must go through debayering and they are rendered as TIFFs. These are then re-rendered together by FFMBC, creating the proxy as specified in the previous menus.



The last tab to go is Folder which points to where each type of file created during the process should be stored, from the original raw files down to the proxies. Keeping these organized will be of great use when the edit is finished and you just need to replace the proxies with DNG sequences.

After filling all these tabs press Start (g) and all the items listed (c) will be transferred to the Processing queue (e) with checkboxes that tell you which files have been processed and which ones are up to go. The Processing log (f) window writes out which step is currently executing and if the previous steps succeeded or failed. It is a good idea to check this once in a while to be sure everything is going as planned.

After all your files go through processing you can hand the proxies to the editor. They have a terribly low quality – plus exposure usually looks pretty weird – but they are enough to check if the shot is in focus or not, for example. It is very important to let the editor know the lens' stretch as well so they can set the proper aspect ratio for the footage in the editing software.

When the edit using the proxies is done, we went to color correction. Our options were DaVinci Resolve or Adobe After Effects through Adobe CameraRaw. Since I was much more familiar with After Effects, this was the package I chose. From version 9.1.5 forward, DaVinci Reslve is fully compatible with MagicLantern's DNGs, not only with ***Cinema DNG*** files.

After bringing my timeline into After Effects, I replaced the proxies with their DNG counterparts. For every clip I replaced a CameraRaw window popped up so I could set the image's parameters. Keeping time as a valuable part of the workflow I made a couple presets to apply to clips that match together, making sure the whole episode had a constant look. This way grading would not have to go crazy to compensate drastic differences between shots.

In the Composition Settings I set the frame size according to the "shooting frame" x "lens stretch" equation we considered before. To stretch every clip I had two possibilities: through the "Interpret Footage" tab on the Project window, or selecting all the clips in the timeline and clicking the option "Transform > Fit To Comp".

After all post-processing it is time to get the final video out of After Effects. We face two alternatives: rendering a "wider than HD" (>1920px width) frame or counter the stretch by decreasing the height of the video instead of increasing its width. Youtube just recently ***started supporting resolutions larger than 1920×1080 pixels***, so stretching your video to 2880x1080px for example, would present a greater risk of losing image quality while streaming. I say this because downscaling from 2880x1080px to 1920x720px would be done by YouTube or Vimeo's converters and not your own settings. Most audiences are happy with Full HD or 720p resolutions these days.

My suggestion is to decrease the height of the clips on the final export and have full control over downscaling instead of leaving it to some internet algorithm. The height percentages for 1.33x, 1.5x and 2x stretch lenses are, respectively, 75,8%, 66.7% and 50%. This keeps our shots 1920px wide but adjusts the height of the frame according to the lenses used. This scenario is particularly useful for high compression codecs since you have a smaller area to fill with information, instead of blowing up compression artifacts over a larger frame.

After all that, our goal was not only test the results online but also on 2k and 4k projectors so I ended up blowing the project to 3072x1152px resolution!

After color correction and effects were done we decided to keep the 2.66:1 aspect ratio instead of traditional CinemaScope. This way the only step left was to export the final version of both episodes. Hitting “Render” translated all this theory into practice!

TITO FERRADANS

"ANAMORPHIC" ON A BUDGET

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